GRID DOOMSDAY FINDER

Constrained quantum-search benchmark for worst-case N-k cascading outages

Solution Design & Architecture — Version 2.0

Event	Q-Hack India 2026 — National Quantum Hackathon (Quantum Week 2026, QuantumRIT, MSRIT Bengaluru)
Track	Track 04 · Open Quantum Innovation
Problem statement	Grid “Doomsday Finder”: quantum search for the worst N-k outage
Core method	Dicke-state / ranked-index Grover–Dürr–Høyer search over exactly-k outage sets, with a deterministic DC/AC quasi-steady-state cascade oracle; QAOA surrogate comparison; structured classical baselines; calibrated fault-tolerant resource and crossover model
Frameworks	Python · NumPy/SciPy/Numba · pandapower / MATPOWER data · Qiskit + Aer · FastAPI + D3
Document version	V2.0 · supersedes V1.0 (6 October 2026)

How to read evidence labels

Every statement about the work carries one of the following status labels wherever it could be mistaken for a result. No quantity in this document labelled DESIGN-STAGE, ESTIMATED, PROPOSED or OPTIONAL is an experimental result.

Label	Meaning
[IMPLEMENTED]	Code exists in the repository, is covered by tests, and can be run by a reviewer.
[MEASURED]	A number produced by an experiment script in `experiments/` from the repository, with its configuration hash recorded.
[SIMULATED]	Produced by running a model of a physical device (e.g. a noise model) on a classical computer.
[EMULATED]	Exact classical computation of the dynamics an ideal quantum algorithm would follow (Tier B). Not a quantum runtime.
[DESIGN-STAGE]	A specification or number derived from printed formulas and stated placeholder parameters; no experiment has produced it.
[ESTIMATED]	An analytical resource estimate. Never a synthesized circuit count unless stated.
[PROPOSED]	Planned work not yet performed.
[OPTIONAL]	Work that will be reported if completed and silently omitted from claims if not.

> **Authenticity note.** This is a design document. It contains no experimental results. Every table cell that must come from an experiment is marked **[to be filled from run]**. Every illustrative number can be regenerated by `resource/cost_model.py` from the parameters printed here.

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1. Executive summary

What problem

Transmission operators must know which small sets of simultaneous branch (line and transformer) outages — N-k contingencies — can trigger a cascade of overload trips and large load loss. For the IEEE 118-bus system (L = 186 branches) there are C(186,3) = 1,055,240 three-branch outage sets and C(186,6) ≈ 5.3×10^{10} six-branch sets. Each set requires a cascade simulation, not a single power-flow solve.

Why it is hard

The number of outage sets grows combinatorially in k; each evaluation is a multi-round, island-aware, nonlinear process; and the objective (unserved load after cascading) is a discontinuous function of the outage set. Exhaustive evaluation is practical for small grids and small k and ceases to be practical at larger k or larger networks (Figure 1 in §2).

What is being built

Grid Doomsday Finder is an open, reproducible pipeline consisting of: (i) a digital twin of IEEE 14-, 30- and 118-bus systems with synthetic, N-1-secure thermal ratings; (ii) a deterministic quasi-steady-state cascade engine in two fidelities — a DC engine used for exhaustive screening and as the search predicate, and an AC engine used for higher-fidelity verification; (iii) a cardinality-constrained quantum-search formulation over exactly-k outage sets using Dicke-state (and ranked-index) encodings with Grover, BBHT and Dürr–Høyer maximum finding; (iv) a three-tier quantum architecture separating circuit demonstration, exact classical emulation and a fault-tolerant oracle design; (v) a synthesized reversible micro-oracle that calibrates the resource model; (vi) strong structured classical baselines; (vii) a QAOA comparison on an explicit quadratic surrogate; (viii) a component-level resource and crossover model; and (ix) an interactive web demonstration.

Why quantum

If the cascade predicate is treated as a black-box oracle over N = C(L,k) candidates, amplitude amplification locates a marked candidate with Θ(√(N/M)) oracle queries, a quadratic reduction relative to the Θ(N/M) queries of unstructured classical search. Whether that query reduction becomes a wall-clock advantage depends on oracle size, logical clock speed, classical parallelism and — critically — how well classical methods exploit the structure of power systems. The project exists to measure and quantify those dependencies, not to assume them away.

What the quantum component demonstrates

- **[IMPLEMENTED/SIMULATED]** Correct circuit-level construction of Dicke preparation, phase oracle and reflection about the Dicke state on toy instances (N = 15–56), executed on Qiskit Aer (ideal and noise-model).
- **[EMULATED]** Exact reduced-space emulation of ideal Grover/Dürr–Høyer dynamics on the full IEEE-scale search spaces, giving oracle-call statistics.
- **[IMPLEMENTED at toy scale]** A synthesized reversible cascade micro-oracle verified by exhaustive basis-state simulation, with counted Toffoli gates.
- **[DESIGN-STAGE / ESTIMATED]** A component-level fault-tolerant oracle model and crossover analysis.

What it does not demonstrate

- No quantum speedup and no quantum advantage. In Tiers A and B the marked set is computed classically and the oracle is a lookup of that result.
- No coherent oracle of IEEE scale has been synthesized; none is feasible on any existing device.
- No claim that the cascade model predicts real blackouts; ratings are synthetic and the model contains no dynamics.
- No claim that Grover outperforms classical heuristics on IEEE grids. We expect structured classical methods to win there, and we report the measurement whichever way it falls.

Practical output

(a) A ranked list of high-severity outage sets under the stated model, flagged by cascade-induced loss, with DC and AC values and an AC non-convergence interval; (b) a measured DC-versus-AC ranking-fidelity study including a false-negative analysis; (c) oracle-call and wall-clock comparisons of amplitude-amplification emulation against exhaustive and structured classical search; (d) a calibrated, parameterised map of the conditions — oracle size, logical clock, classical parallelism and classical scaling exponent — under which a future fault-tolerant quantum oracle could become competitive.

2. Problem, motivation, objectives and research questions

2.1 Problem statement

Given a transmission network with L candidate branches, a stressed N-1-secure base operating point and a deterministic cascade model, find the exactly-k simultaneous initiating outages that maximize unserved load when the cascade terminates, and characterize the cost of finding them by classical and quantum search.

2.2 Motivation

- **Operational relevance.** N-1 security is routine. Planning standards such as NERC TPL-001 formalise N-1 and selected multiple-contingency events; exhaustive N-k screening for k ≥ 3 is not performed in practice because of the combinatorics.
- **Multi-element events.** Major disturbances such as the 30–31 July 2012 grid disturbances in India involved several elements out of service, heavy inter-regional flows and subsequent protection operations. Figures about affected population must be cited from the Ministry of Power / CEA enquiry report before they are shown on a slide [PROPOSED: citation check].
- **Quantum-computing relevance.** Worst-case contingency search is a constrained combinatorial search with a checkable predicate, a canonical target for amplitude amplification. It is also an instructive test case for the gap between query complexity and wall-clock advantage.

2.3 The combinatorial wall

Grid	L	k=2	k=3	k=4	k=5	k=6
IEEE-14	20	190	1,140	4,845	15,504	38,760
IEEE-30	41	820	10,660	101,270	749,398	4,496,388
IEEE-118	186	17,205	1,055,240	48,277,230	1.757×10^{9}	5.30×10^{10}
Polish-2383 (resource-model proxy only)	≈2,896	$\approx4.2\times10^{6}$	$\approx4.0\times10^{9}$	$\approx2.9\times10^{12}$	—	—

Table 2.1 — Number of outage sets N = C(L,k). Values are exact binomial coefficients (Polish-2383 branch count approximate).

At an illustrative 10 ms per AC cascade evaluation on one core, IEEE-118 at k = 3 costs ≈ 2.9 hours, k = 4 ≈ 5.6 days, and k = 5 ≈ 203 days [DESIGN-STAGE: the 10 ms figure is a placeholder to be replaced by E1]. Parallel hardware reduces these linearly; growth in k is exponential.

2.4 Objectives

- O1. Define a mathematically precise exactly-k worst-case cascade search problem with explicit metrics and terminology.
- O2. Build and validate a deterministic quasi-steady-state DC/AC cascade oracle and measure DC-versus-AC ranking fidelity.
- O3. Implement and verify cardinality-constrained amplitude-amplification machinery (Dicke/ranked index, Grover, BBHT, Dürr–Høyer) at circuit level on toy instances and by exact emulation at IEEE scale.
- O4. Synthesize a toy-scale reversible cascade oracle and use it to calibrate a component-level resource model.
- O5. Compare against strong, structure-exploiting classical baselines under a fair protocol and report the outcome honestly.
- O6. Quantify the conditions under which a quadratic query reduction becomes a wall-clock advantage.
- O7. Deliver an interactive demonstration that separates measured, emulated, modelled and proposed content.

2.5 Research questions

ID	Question	Answered by
RQ1	Which exactly-k outage sets are most severe under the model, and how many are cascade-driven rather than simple isolation?	E2
RQ2	How faithfully does a DC cascade screen reproduce AC-verified severity labels and rankings, and how often does it miss sets that AC finds severe?	E3

ID	Question	Answered by
RQ3	Are the findings properties of the grid and cascade rules, or artefacts of the synthetic rating rule?	E11
RQ4	Do the circuit constructions (Dicke, phase oracle, reflection) and the exact emulation agree, and do GAS-style searches follow the predicted √N scaling?	E4–E6
RQ5	Does amplitude amplification outperform reasonable classical alternatives in oracle evaluations on IEEE grids?	E7
RQ6	Under what oracle size, clock speed, parallelism and classical scaling exponent does a quadratic query reduction yield a wall-clock advantage?	E14, E15
RQ7	What is the surrogate gap between a quadratic QAOA objective and the true cascade severity?	E10

3. System scope and assumptions

3.1 Scope statement

> **Scope statement (verbatim for slides and README).** We model a quasi-steady-state cascading-outage process. After the initiating outage, we repeatedly identify electrical islands, rebalance each island by redispatching generation within its limits and curtailing load pro-rata if a deficit remains, solve power flow, and trip all branches whose loading exceeds a fixed multiple of their rating. The process ends when no branch is overloaded, when thirty rounds are reached (flagged), or when no load is served. We do not model dynamics, frequency, protection timing, hidden failures, operator action or remedial schemes. The output, the unserved-load fraction, is a screening severity metric for comparing outage sets within this model. It is not a prediction of real blackouts.

3.2 Three levels of evidence

Level	Role	What it supports	What it does not support
DC model	Exhaustive screening / reference model and the search predicate	Exact reference severities *for the DC model*, enabling exact validation of search algorithms; ranking fidelity once measured against AC (E3)	Voltages, reactive flows, losses, MVA limits, or any statement about physical reality
AC model	Higher-fidelity static verification	Whether DC-identified sets remain severe when voltages, losses and MVA limits are included, under identical cascade rules	Dynamics, protection timing, operator response, real-grid behaviour
Real grid	Future validation target	Nothing is claimed	Requires utility models, real ratings and dynamic studies; real data are often access-restricted

3.3 Principal assumptions

#	Assumption	Consequence / treatment
A1	Initiating outages are simultaneous and instantaneous	A screening convention for coordinated or common-cause events; not a timeline of real cascades
A2	Ratings are synthetic: $R_l = \max((1+h)\cdot\lvert flow_l^0\rvert, s_{floor})$	Findings characterise the model; robustness tested over rating families (E11)
A3	Overloads trip instantly at a multiple of rating; no time-overload curve	Pickup-margin sweep $\tau_{trip} \in \{1.0, 1.1, 1.2, 1.25\}$
A4	Generation redispatches pro-rata within limits; loads are constant-power; curtailment is uniform within an island	Optimistic abstraction; no-upward-redispatch variant brackets it
A5	Exactly-k outages (fixed Hamming weight)	S is not monotone in the outage set; ≤k is an extension handled by a k-sweep (§4.4)
A6	The quantum predicate is the DC cascade	A coherent oracle can plausibly compute only DC-type linear algebra; AC is verification

4. Mathematical problem formulation

4.1 Search space

Let branches be indexed 1,...,L. An initiating outage set is represented by $x \in \{0,1\}^L$ with $x_i = 1$ iff branch i is lost. The exactly-k space is

$$\Omega_k = \{ x \in \{0,1\}^L : |x| = k \},\ N = |\Omega_k| = C(L,k).$$

Each $x \in \Omega_k$ corresponds to exactly one k-subset of branches, so N is exactly the binomial coefficient. Parallel branches between the same buses are separate branches and separate bits.

4.2 Severity and cascade-induced loss

Let Cascade(x) be the deterministic process of §8. Define the severity $S(x) \in [0,1]$ as the fraction of base-case load unserved at termination, $S_0(x)$ as the fraction unserved immediately after the initial outage, island balancing and curtailment of round 0 and before any overload trip, and

$$C(x) = S(x) - S_0(x) \geq 0 \text{ (cascade-induced loss).}$$

S_0 measures direct isolation (e.g. loss of every branch to a load bus); C measures loss created by the overload cascade. Diagnostic ratio $A(x) = S/\max(S_0, \varepsilon)$, $\varepsilon = 0.01$, is displayed but is not used for ranking because it is unstable when $S_0 \approx 0$.

4.3 Problems

ID	Problem	Definition	Algorithm (§12)
A	Worst-set search (headline)	$x^* = \text{argmax } S(x)$ over $x \in \Omega_k$. Success means returning any x with $S(x) \geq S_{max} - 1e\text{-}9$ (ties are common)	Dürr–Høyer maximum finding with BBHT subroutine
B	Threshold search	Find any x with $S(x) \geq \tau$, i.e. $f_\tau(x) = 1$	BBHT with oracle f_τ
C	Top-T ranking	Return the T highest-severity sets	Repeated maximum finding with exclusion list; $\approx 2\sqrt{(NT)}$ queries
D	Resource / crossover	For which N, oracle cost, quantum clock, parallelism and classical exponent is $T_{quantum} < T_{classical}$?	Resource model (§16)
E	Cascade-driven high-severity sets	$S(x) \geq \tau$ and $C(x) \geq \delta$	Post-filter on A–C outputs

4.4 Why exactly-k, and why not ≤k

Exactly-k fixes the Hamming weight, so the valid space is a single Dicke subspace and the search register has an exact uniform initial state (§11). S is **not monotone**: adding an outage can reduce S, for example by isolating a stressed region so that less load is shed. Consequently the worst ≤k set need not be a worst k-set. The primary problem is exactly-k; ≤k is an extension handled either by a k-sweep or by a superposition over Dicke states of weights 0..k, at modest additional preparation cost. Experiment E2 measures the fraction of nested pairs $(x, x \cup \{j\})$ for which S decreases.

4.5 Oracle contract

All search, baseline and resource code interacts with the grid only through a pure function $S: \Omega_k \to [0,1]$ (batched), with derived predicates $f_\tau(x) = [S(x) \geq \tau]$ and $f_y(x) = [S(x) > y]$. Strict inequality in the adaptive form guarantees termination in the presence of ties. The oracle is a pure function of the *set* x (order-invariant) and is deterministic to a stated tolerance (§8.8).

5. Power-grid digital twin

5.1 Data and the internal Grid object [PROPOSED → IMPLEMENTED in sprint]

- **Cases.** IEEE 14-, 30- and 118-bus systems from pandapower's network library or MATPOWER case files. Source and version are recorded in every result file.
- **Internal Grid dataclass.** Bus table (id, type, P_d, Q_d, V_{min}/V_{max}), branch table (from, to, r, x, b, tap, status, rating), generator table (bus, P_{max}, P_{min}, P_g, V_{set}), base MVA. Loaders convert library objects into this structure so that the cascade engine has no dependency on a library's object model.
- **Candidate set.** All branches (lines and transformers) by default, with a type tag so results can exclude transformers. L = |C| equals the number of decision bits.
- **Parallel branches** are independent bits. Reports distinguish "one of two parallel lines" from "both".

5.2 Base operating point

The base operating point is defined in a fixed order so that DC and AC views derive from one physical state:

- **Step 1.** Load the case; all branches in service.
- **Step 2.** Apply stress $\lambda \geq 1$: loads and generation scaled pro-rata within generator limits.
- **Step 3.** Solve AC Newton–Raphson (tolerance 10^{-8} p.u., ≤ 30 iterations). Require convergence and bus voltages within [0.90, 1.10] p.u.
- **Step 4.** Derive ratings from this single AC solution (§5.3).
- **Step 5.** Set DC base-case injections equal to the AC base-case bus active-power injections (including the slack's loss pickup). DC and AC then share a dispatch, and their base flows differ only through voltage and angle effects.
- **Step 6 (secure-base definition).** No branch exceeds its rating at the base point, and every *non-bridge* single outage settles with zero overload trips, checked per engine. Bridge outages (radial isolation) necessarily have $S_0 > 0$; they are tagged `bridge`, reported, and not counted as N-1 failures.

5.3 Thermal-rating synthesis

For each in-service branch l with base-case AC apparent flow $|S_l^0|$ (MVA) and active flow $|P_l^0|$ (MW):

$$R_l^{AC} = max((1+h)\cdot|S_l^0| , s_{floor}),\ R_l^{DC} = max((1+h)\cdot|P_l^0| , s_{floor}),\ h \in [0.15, 0.40].$$

s_{floor} (default 5% of the largest base flow) prevents near-zero-flow lines from being trivially fragile. Deriving both ratings from one AC base point removes the unit mismatch that would arise if DC active-power flows were compared with AC apparent-power ratings. Calibration (`calibrate.py`) sweeps (h, λ) and selects a setting that satisfies the secure-base definition while producing a non-trivial fraction of N-2/N-3 cascades. **The ratings are synthetic.** They are not utility data, and the fraction of vulnerable sets is partly a property of the rating rule. Experiment E11 tests this directly, including rating families not proportional to base flow (nominal MVA classes by voltage level; case-file ratings where non-zero; ±10% random perturbation).

5.4 Contingency generation

- Enumeration of Ω_k in lexicographic order of sorted branch indices, with **ranking and unranking** in the combinatorial number system: $r \leftrightarrow (c_k > \ldots > c_1)$ with $r = \Sigma\ C(c_i, i)$. This provides random access to the r-th outage set, supports chunked and parallel enumeration, and is the index used by the ranked-index quantum encoding (§11.4).
- Batch interface: an outage matrix of shape (n, k) of sorted branch indices.
- Structural reductions that preserve exactness (bridge/radial-spur handling in closed form) are optional and always labelled; heuristic shortlists are used only to fit toy circuits on a device and are never evidence of scaling.

6. DC power-flow model

6.1 Model and assumptions

The DC model linearises AC power flow under flat voltage magnitudes, small angle differences and lossless branches. With branch susceptance $b_l = 1/x_l$ and branch-bus incidence matrix A (L×B):

$$B' \theta = P,\ f = diag(b)\ A\ \theta,\ B' = A^T\ diag(b)\ A \text{ (slack row/column removed).}$$

It discards voltage magnitudes, reactive power, losses and MVA limits. It is appropriate for exhaustive screening, which is its standard role in contingency analysis and N-k interdiction literature, and it is the only part of the cascade whose arithmetic a coherent quantum oracle could plausibly evaluate. It is **not** a ground truth for the physical grid, and this document never calls it one.

6.2 Sensitivities and the compensation (Woodbury) formula

Define the L×L branch-to-branch sensitivity matrix $\Phi = diag(b)\ A\ B'^{-1}\ A^T$; $\Phi_{l,m}$ is the flow change on branch l per unit transfer across the terminals of branch m. For an outage set K, the post-outage flows on surviving branches $l \notin K$ are

$$f'_l = f_l + \Phi_{l,K}\ (I - \Phi_{K,K})^{-1}\ f_K,\ f'_K = 0.$$

For $K = \{m\}$ this is the familiar $f'_l = f_l + LODF_{l,m}\ f_m$ with $LODF_{l,m} = \Phi_{l,m}/(1 - \Phi_{m,m})$. Applying the rank-|K| update costs $O(L|K| + |K|^3)$ per set after precomputing Φ, versus a fresh factorisation of B′; it is vectorisable over many outage sets. The derivation is in Appendix C.3.

> **Islanding rule.** The compensation formula holds at fixed topology and fixed injections. Island detection is therefore performed by **exact graph connectivity** before any Woodbury solve; it is never inferred from a condition-number threshold. Whenever an outage set (or a later trip) disconnects the network, each island is solved separately with its own slack, rebalanced injections and its own B′ reduced to the island. Test U5 verifies that $\Phi_{K,K}$ singularity coincides with disconnection on random K.

6.3 Performance targets [DESIGN-STAGE; to be replaced by E1]

Engine	Grid	Target per evaluation	Method
DC cascade (non-islanding sets)	IEEE-118	≲ 0.5 ms single core; batched ≲ 0.05 ms	Precomputed Φ, Woodbury update, NumPy/Numba, chunked batches
DC cascade (islanding sets)	IEEE-118	Slower; timed separately	Explicit per-island solve
AC cascade	IEEE-118	≲ 10–30 ms	Sparse Newton–Raphson with warm starts
DC reference table	IEEE-118, k = 3 (1.06×10^6 sets)	≲ 20 min laptop	Chunked, cached as float32 severity + bitmask

These are targets. Every classical time used in the resource model is replaced by the measured mean and p95 from E1, split by class (non-islanding, islanding, long cascade).

7. AC power-flow model

Bus power balance with complex voltages $V_i = |V_i|\angle\delta_i$ and admittance matrix Y:

$$S_i = V_i \cdot \mathrm{conj}\left(\Sigma_j Y_{ij} V_j \right) = P_i + jQ_i \text{ for every bus } i.$$

The system is solved by Newton–Raphson with PV/PQ/slack bus types, independently per island; the slack of each island is its largest-P_{max} in-service generator. Generator reactive limits (PV→PQ switching) are an advanced option, **off by default** for speed and reproducibility; with limits off, voltage-driven infeasibility is rarer and AC non-convergence is more often numerical than physical. This is disclosed in all AC results. Branch flows S_l come from the converged voltages.

- **Role.** Higher-fidelity static verification of reported sets, DC-versus-AC agreement studies (E3), and full AC reference tables wherever exhaustive evaluation is tractable (IEEE-14 k ≤ 4, IEEE-30 k ≤ 3, IEEE-118 k ≤ 3 at roughly 10^6 evaluations). Beyond that, AC is applied to the top-T sets and to stratified samples.
- **Not a coherent oracle.** AC power flow is never priced as a quantum oracle.

8. Cascade simulation

8.1 Algorithm

```
INPUT : Grid G, base state, outage set x (|x| = k), mode in {DC, AC}, config
        (tau_trip = 1.0, policy = simultaneous, R_max = 30, ladder_step = 0.05)
OUTPUT: S, S0, C, rounds, trips_per_round, islands, status, flags, frames

 1  status[l] <- IN_SERVICE for all l;  status[x] <- OUT
 2  for r = 0 .. R_max - 1:
 3      islands <- connected_components(in-service branches)            # exact graph search
 4      for each island I:
 5          if no generator in I:      served[I] <- 0 ; continue         # dead island
 6          balance(I)               # redispatch within [Pmin,Pmax]; uniform load curtailment if deficit
 7          (ok, flows[I]) <- solve_flow(I, mode)       # DC: Woodbury / reduced B' solve; AC: Newton-Raphson
 8          if not ok:  (AC only)    (ok, flows[I]) <- curtailment_ladder(I)   # see 8.6; flag recorded
 9          served[I] <- served_load(I)
10      u[l] <- |flow[l]| / rating[l]        for in-service l      # MW/MW (DC) or MVA/MVA (AC)
11      O <- { l : u[l] > tau_trip + 1e-9 }
12      if r == 0: S0 <- 1 - sum(served)/initial_load
13      record(r, status, flows, islands, served, O)
14      if sum(served) == 0:  status_out <- TOTAL_LOSS ; break
15      if O is empty:        status_out <- STABLE     ; break
16      trip(O, policy)       # simultaneous: all of O ; sequential: only argmax u (ties: lowest index)
17  else: status_out <- EXHAUSTED                                    # flagged, never silent
18  S <- 1 - sum(served)/initial_load ;  C <- S - S0
19  return S, S0, C, rounds, status_out, flags, frames
```

8.2 Step-by-step explanation

- **Initial outage (step 1).** The k branches are removed instantaneously; generation and load are unchanged until island balancing.
- **Islands (step 3).** Connected components of the in-service graph. Connectivity is authoritative.
- **Balance (steps 5–6).** No generator: all island load is lost and no flow is solved. Otherwise let $\Delta = \Sigma P_{gen} - \Sigma P_{load}$. Surplus: scale generation down pro-rata (flagged `pmin_violation` if P_{min} would be violated). Deficit: first raise generation pro-rata to headroom up to P_{max}; any remaining deficit is met by uniform pro-rata load curtailment. A no-upward-redispatch variant is a sensitivity case.
- **Power flow (step 7).** DC or AC per island. Loading $u_l = |\text{flow}_l|/R_l$ in consistent units (steps 10).
- **Overload set (step 11).** Strict inequality with an explicit tolerance of 10^{-9}.
- **Trip policy (step 16).** Primary: all overloaded branches trip in the same round (deterministic and vectorisable). Variant: only the single most-overloaded branch trips per round, giving finer timelines. The policy is recorded in every result.
- **S_0 capture (step 12).** Round-0 served load, after balancing and before any trip, defines S_0 so $C = S - S_0$ isolates cascade-induced loss.

8.3 Stopping states

State	Condition	Treatment
STABLE	No overloaded branch	Normal termination
EXHAUSTED	$R_{max} = 30$ reached with trips remaining	Flagged; last state is the output; fraction reported
TOTAL_LOSS	No load served	Terminal; $S = 1$
STATIC_INFEASIBLE	An island entered the AC non-convergence ladder (§8.6)	Flagged with ladder step and interval

8.4 Edge cases

Case	Handling
Island without generator	Entire island load unserved; no flow solve
Island generation shortfall / surplus	Redispatch within limits, then pro-rata curtailment / generation reduction
Island without slack (AC)	Largest-P_{max} in-service generator becomes slack
Parallel branches	Independent branches and independent trip logic
Near-singular $I - \Phi_{KK}$	Never divided through; connectivity decides, explicit island handling follows
Transformers	Same branch model with tap ratio; DC treats as reactance only
Numerical noise near rating	Strict greater-than with 10^{-9} tolerance; documented
Ties in S	Equality band $\lvert\Delta S\rvert < 10^{-9}$; success defined against S_{max} with this band

8.5 What the model does and does not simulate

Simulated (quasi-steady state)	Not simulated
Branch removal; island formation; per-island balance; static power flow; thermal overload tripping at a fixed multiple of rating; repeated rounds	Transient and frequency dynamics; generator and governor dynamics; relay coordination and timing; hidden failures; operator actions and remedial schemes; load dynamics; under-frequency and voltage protection; full voltage-stability dynamics; economic dispatch/OPF

8.6 AC non-convergence handling

Non-convergence can indicate that no solution exists at the island's loading or that the numerical method failed; the model does not claim to distinguish them. The rule is deterministic:

- Run Newton–Raphson from a flat start; on failure, retry once from DC-angle initialisation.
- If both fail, flag the island `STATIC_INFEASIBLE` at its current loading.
- Apply the **curtailment ladder**: scale island load (P and Q together) by $(1 - 0.05\,j)$, $j = 1,\ldots,20$, and accept the first j that converges. If none converges, de-energize the island.
- Every result records `nonconv_flag`, `ladder_step` and the bounding interval $[S_{ladder}, S_{total}]$, where S_{total} treats the failed island as fully lost. Rankings use S_{ladder}; E11 reports how often the two disagree and whether the top-10 changes.
- The fraction of reported sets that depend on the ladder appears in every results table.

> **Model statement (verbatim).** If the AC power flow of an island fails to converge from both flat and DC-initialized starts, we flag the island as statically infeasible at its current loading. As a numerical proxy for emergency load reduction, island load is then scaled down uniformly in 5% steps until a solution is found; if none is found the island is de-energized. This is a modelling abstraction, not a representation of any utility's protection scheme or of a demonstrated voltage collapse. Every affected result is flagged, and results are reported both with the ladder and with total island loss as a bounding case.

8.7 Parameters and variants

Parameter	Default	Variants (sensitivity)
Trip factor τ_{trip}	1.0	1.0, 1.1, 1.2, 1.25
Trip policy	all-overloaded	most-overloaded-first
R_{max}	30	5, 10, 30
Redispatch	upward + downward	no upward redispatch
AC tolerance	10^{-8} p.u.	10^{-6}, 10^{-10}
Ladder step	5%	1%, 5%, 10%, immediate total loss

8.8 Determinism

Branch order is fixed by the case file; sorts use (value, index); floating-point tolerances are explicit constants; float64 throughout; a single-threaded reference mode exists. Reproducibility is claimed **to tolerance (10^{-9})**, not bit-identity across machines, because BLAS thread reductions can reorder floating-point operations. The oracle must be a pure function of x for amplitude amplification to be well defined, and reviewers must be able to rerun the results.

9. Severity metrics

Metric	Role	Rationale
S(x) — MW-weighted unserved load fraction	**Primary** objective and oracle output	Bounded in [0,1], comparable across grids, operator-meaningful under a static model
S_0(x) — loss before any overload trip	Secondary	Separates direct isolation from cascade
C(x) = S − S_0	Secondary; preferred cascade-driven indicator	Stable when $S_0 \approx 0$
A(x) = S / max(S_0, ε)	Diagnostic only	Ratio unstable near $S_0 \approx 0$
Rounds, #trips, #islands, generation lost (MW), ladder/exhausted flags	Diagnostic	Describe cascade shape
Weighted (priority) load loss	Not used	Weights would be arbitrary
“Unserved energy” (MWh)	Not used	No time axis; the term would be incorrect

9.1 Terminology

Term	Use in this project
Contingency	Initiating loss of a specified set of branches
N-k contingency	A contingency of exactly k simultaneous branch outages
Cascade / cascading outage	In this model: the sequence of overload-triggered trip rounds after the initiating contingency, in quasi-steady state with no time axis
Load shedding	Only as “modelled load curtailment during island balancing”; never “protection operation”
Collapse	Avoided; use “static infeasibility flag”. “Voltage collapse” appears only as “a possible indicator of voltage instability”
Blackout	Not a technical term here. Pitch-only, always qualified as “blackout-scale load loss in our model”; never “blackout prediction”
Catastrophic	Not used in technical text; use “high-severity (S ≥ τ)”
Dangerous outage set	“High-severity outage set under this model”; “cascade-driven high-severity set” if S ≥ τ and C ≥ δ

Default τ = 0.20 of base load and δ = 0.10 are calibration parameters, reported with every table, and swept in E8.

10. DC/AC fidelity strategy

DC is useful only to the extent that it ranks outage sets as AC does. The project therefore treats DC–AC agreement as a **measured result**, not a design assumption. The dangerous error direction is the false negative: a set that DC calls benign and AC calls severe.

10.1 Experiments establishing whether DC screening is trustworthy

Exp.	Design	Metrics
E3a Exhaustive	Full AC and DC tables: IEEE-14 ($k \le 4$), IEEE-30 ($k \le 3$), IEEE-118 ($k \le 3$)	Label agreement at τ (precision, recall, F1); Spearman ρ on the full ranking; top-10/100 overlap and Jaccard; S-error distribution; **false-negative rate**
E3b Sampled	IEEE-118 $k = 4$: AC evaluated on a sample stratified over DC-severity deciles	Recall by decile (verifying only DC top sets cannot measure recall)
E3c Decomposition	(i) common ratings from one AC base point (primary); (ii) per-engine ratings; (iii) AC with ladder disabled	Separates rating-derivation effects, physics effects and ladder effects

> **Pre-registered decision rule.** Thresholds are fixed before the experiment is run. DC screening is called "trustworthy for this model" only if recall at $\tau \ge 0.9$ and top-10 overlap ≥ 0.5. Otherwise the finding is reported as "DC is a poor proxy here", and the DC-to-AC fidelity ladder becomes a result of the study rather than a design assumption. Thresholds may be changed only before the data are inspected.

11. Quantum search formulation

11.1 The Hamming-weight search space

Start from $x \in \{0,1\}^L$ with $|x| = k$. The valid configurations are exactly the k-subsets of the L branches, so $N = C(L,k)$, and every one of them is a legitimate initiating outage. Strings of any other weight are not outage sets of the problem and must never be amplified or measured as candidates. A register of L qubits spans 2^L basis states, of which only N are valid; the encodings below restrict the search to exactly those N.

11.2 The Dicke state

$$|D_k^L\rangle = C(L,k)^{-1/2} \sum_{x\in\{0,1\}^L,\ |x|=k} |x\rangle.$$

- **Normalisation.** $\langle D|D\rangle = \Sigma_x\ (1/N)\cdot 1 = N\cdot(1/N) = 1$ because the N basis states are orthonormal and each carries amplitude $N^{-1/2}$.
- **Uniform on the valid space.** Let $H_k = \text{span}\{|x\rangle : |x| = k\}$, $\dim H_k = N$. In the index basis, $|D\rangle$ is the uniform vector $(1,\ldots,1)/\sqrt{N}$.
- **Closure.** The phase oracle is diagonal in the computational basis, so it maps each weight-k basis state to itself times ±1 and preserves H_k. The reflection $2|D\rangle\langle D| - I$ maps H_k to itself because $|D\rangle \in H_k$. The state never leaves H_k.
- **Complexity.** Grover amplification inside an N-dimensional space with M marked states needs $\lfloor \pi/(4\theta) \rfloor \approx (\pi/4)\sqrt{(N/M)}$ iterations with $\sin\theta = \sqrt{(M/N)}$, i.e. **$O(\sqrt{C(L,k)})$ oracle queries**.

> **Careful comparison.** Starting from the full uniform state over 2^L strings and filtering weight inside the oracle would cost a factor $\sqrt{(2^L/C(L,k))}$ more, but that is not the relevant alternative. The relevant alternatives are other exactly-N encodings (ranked index, §11.4) that achieve the same query complexity. The Dicke encoding is chosen for the circuit tiers for its simplicity and its direct physical meaning, not because it is universally superior.

11.3 Dicke-state preparation and its cost

The Bärtschi–Eidenbenz construction prepares $|D_k^L\rangle$ deterministically and exactly from $|0\ldots0\ 1^k\rangle$ with O(kL) gates and O(L) depth, using cascaded two- and three-qubit "split-and-cyclic-shift" blocks of controlled-R_Y and CNOT gates. The implementation is unit-tested against the analytic state for all $L \le 10$, all k (Test Q1), with Qiskit state preparation from an explicit amplitude vector as the exact fallback for toy sizes.

Each Grover iteration uses the preparation twice (U and $U^\dagger$) inside the reflection. In the resource model the cost is $C_D \approx k\cdot L\cdot c_{rot}$ Toffoli-equivalents with $c_{rot} \approx 10$–30 covering synthesized controlled rotations. This is a parameter calibrated against transpiled gate counts (E4/E14), not assumed to be zero. **[DESIGN-STAGE]** For IEEE-118 at $k = 3$ and $c_{rot} = 20$, $2C_D \approx 2\times558\times20 \approx 2\times10^4$ Toffoli-equivalents, about 0.05% of the realistic oracle cost per iteration and about 0.5% of the optimistic one. Conclusion: Dicke preparation does not change the practical argument, and it remains an explicit term in the model.

11.4 Ranked-index alternative

Encode the outage set by its rank $r \in \{0,\ldots,N-1\}$ in the combinatorial number system, using $\lceil \log_2 N \rceil$ qubits ($\approx$ 21 for IEEE-118 at $k = 3$ versus 186 for the Dicke register). The initial state is the uniform superposition over $[0,N)$ (prepared with standard non-power-of-two techniques); the diffuser reflects about it. The oracle first *unranks* r into a sorted k-index list using a binomial-coefficient QROM, at an estimated O(kL log L) Toffolis per call — small against the flow arithmetic. The Grover problem is identical (N states, no duplicates), so query complexity is unchanged.

Encoding	Decision qubits	States searched	Iteration factor	Role
Binary + Dicke	L (186 for IEEE-118)	N	1	Circuit tiers (A, B); cleanest physical picture
Ranked index	$\lceil \log_2 N \rceil$ ($\approx$ 21 for IEEE-118, $k = 3$)	N	1	Tier C default; unranking cost included in the oracle
k index registers (naive)	$k\lceil \log_2 L \rceil$ (24 qubits for IEEE-118, $k = 3$)	L^k with duplicates and orderings	$\approx \sqrt{(L^k/N)}$ (≈ 2.5 at $k = 3$; ≈ 5.0 at $k = 4$)	Reported only as a rejected alternative

12. Search algorithms

Four related problems are kept distinct; each has its own algorithm and query-complexity statement.

Problem	Algorithm	Oracle	Query count	Notes
1. One marked set, M known	Grover with $r^* = \lfloor \pi/(4\theta) \rfloor$, $\sin\theta = \sqrt{(M/N)}$	$f(x) = [x \in M]$	$\approx (\pi/4)\sqrt{(N/M)}$	Toy and validation runs only
2. One marked set, M unknown	BBHT: iteration count j uniform in {0,...,m−1}, m grown by $\lambda = 6/5$ up to $\sqrt{N}$	same	Expected $O(\sqrt{(N/M)})$; explicit constant per Boyer–Brassard–Høyer–Tapp	Classical check after each measurement
3. Any set with $S \geq \tau$	BBHT with f_τ. Quantum counting would estimate M; here M is table-derived (Tier B) or modelled (Tier C)	$f_\tau = [S \geq \tau]$	$O(\sqrt{(N/M)})$	Random sampling needs $\approx N/(M+1)$: when M/N is large, classical wins (E8 maps this regime)
4. Maximum S	**Dürr–Høyer maximum finding with BBHT subroutine (GAS-style threshold ascent)**: threshold y, mark $S > y$, raise y on success, stop when the schedule is exhausted	$f_y = [S > y]$ (strict)	Proven $E \leq 22.5\sqrt{N} + 1.4 \lg^2 N$ (Dürr–Høyer); empirical c measured in E6	Headline run $y_0 = 0$ (pure quantum count); secondary run y_0 from a classical heuristic
5. Top-T ranking	Repeated maximum finding with an exclusion list (or threshold ladder)	f marks $S > y$ and $x \notin$ found-list	$\approx \Sigma_{j \leq T} \sqrt{(N/j)} \approx 2\sqrt{(NT)}$ [DESIGN-STAGE; E9]	Exclusion list adds T·k comparators to the oracle; reported separately

12.1 Grover iteration geometry

Let $|a\rangle$ and $|b\rangle$ be the normalised uniform superpositions over marked and unmarked states, so $|s\rangle = \sin\theta\,|a\rangle + \cos\theta\,|b\rangle$ with $\sin^2\theta = M/N$. The oracle is the reflection $I - 2|a\rangle\langle a|$ and the diffuser $2|s\rangle\langle s| - I$; their product rotates by 2θ. After r iterations

$$P_{success}(r) = \sin^2((2r+1)\theta),\ r^* \approx \pi/(4\theta) - ½ \approx (\pi/4)\sqrt{(N/M)}.$$

Worked numbers [exact arithmetic]: N = 1,140 (IEEE-14, k = 3), M = 1: $\theta = 0.02962$ rad, $r^* \approx 26$. N = 1,055,240 (IEEE-118, k = 3), M = 1: $r^* \approx 806$. Toy N = 15 (m = 6 branches, k = 2), M = 1: $\theta = 0.2612$ rad; r = 2 gives P = 0.933 and r = 3 gives P = 0.935.

12.2 Dürr–Høyer maximum finding with BBHT (specification)

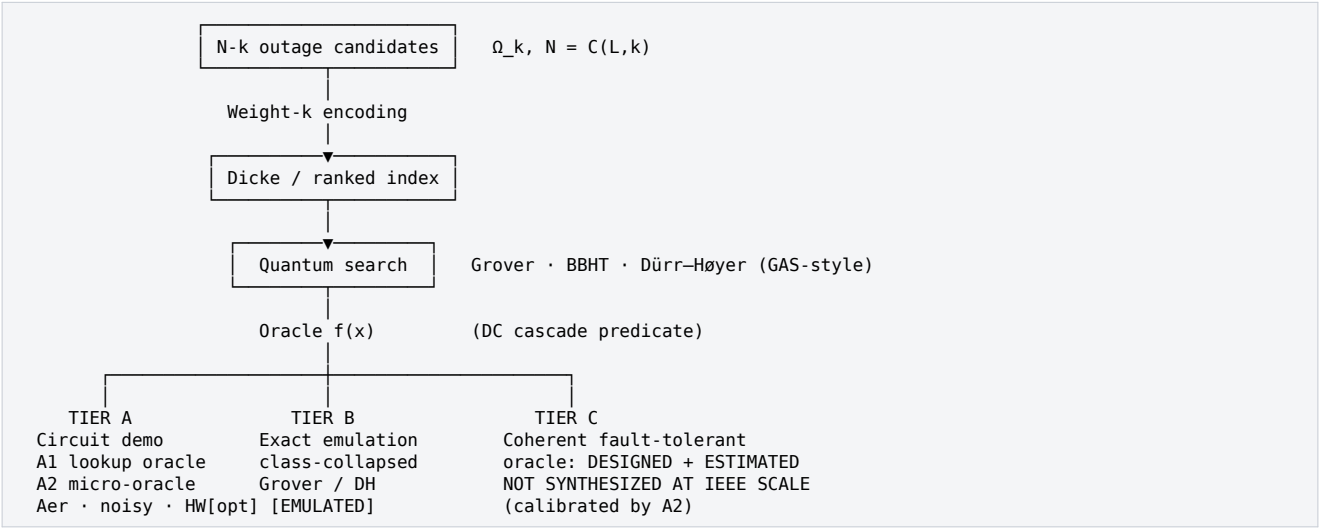
```
def dh_max(oracle_table_or_fn, N, y0=0.0, lam=6/5, m_cap=sqrt(N), budget, rng, patience=3):
    y, best, calls, fails = y0, None, 0, 0
    while calls < budget and fails < patience:
        m, found = 1.0, False
        while m <= m_cap and calls < budget:
            j = rng.integers(0, ceil(m))                # random iteration count
            x = measure_after(j, mark = (S > y))        # j Grover iterations, then measure
            calls += j + 1                              # quantum oracle calls (+1 for the marking check)
            verifications += 1                          # one classical evaluation, reported separately
            if S[x] > y: y, best, found = S[x], x, True ; break   # raise threshold, restart schedule
            m *= lam
        fails = 0 if found else fails + 1              # exhausted schedule without improvement
    return best, y, calls, verifications
```

The m cap, the restart rule and the termination rule (m schedule exhausted with no improvement for *patience* consecutive attempts, default 3) are specified once and implemented once; the failure probability is tied to the patience parameter and estimated empirically (E6), not asserted as “with high probability”. Quantum oracle calls are counted conservatively as $\Sigma(j+1)$ per attempt; classical verifications are separate real evaluations.

12.3 Reporting convention (all methods)

- **Discovery calls**: unique evaluations until S_{max} is first observed.
- **Certified-stop calls**: evaluations until a principled stopping rule fires with stated confidence. Only Dürr–Høyer and brute force have one; heuristics do not, and the comparison states this.
- Success is defined against S_{max} with the 10^{-9} tie band, never against a particular x.

13. Quantum oracle architecture



Tier	What runs	What it proves	What it does NOT prove	Status
A1	Dicke prep, compiled phase oracle, reflection, measurement; m = 6–8 candidate branches, k = 2–3 (N = 15–56); Aer ideal, Aer noise model, optional IBM device	Circuit construction is correct; ideal amplification matches theory; noise degrades it as expected	Any speedup (the marked set is computed classically); scaling; hardware advantage	[IMPLEMENTED] on simulators; hardware [OPTIONAL]
A2	Synthesized reversible cascade micro-oracle on a toy grid	A compute–phase–uncompute cascade oracle compiles and uncomputes cleanly with counted Toffolis; calibrates the cost model	That an IEEE-scale oracle is realizable	[IMPLEMENTED] at toy scale
B	Exact classical emulation of ideal Grover / Dürr–Høyer dynamics in the weight-k subspace on full IEEE spaces	Search logic, oracle-call statistics and success rates for a given predicate table; equivalence with the circuit (Test Q4)	Oracle realizability, noise, wall-clock time or hardware behaviour	[EMULATED]
C	Analytical component-level cost model of a coherent fault-tolerant oracle	Conditions under which a quadratic query reduction could be a wall-clock advantage	Anything physical; no circuit of this size has been synthesized or run	[DESIGN-STAGE] [ESTIMATED]

13.1 Tier A — circuit demonstration

- **Dicke preparation** (§11.3), validated against the analytic state.
- **Phase oracle (A1).** A classical computation of the marked set $\mathcal{M} = \{x : S(x) \geq \tau\}$ on the candidate subset; the circuit applies, for each $x \in \mathcal{M}$, a multi-controlled Z conjugated by X gates on the qubits where x has a 0. Cost $O(|\mathcal{M}|\cdot m)$ multi-controlled gates. **The marked set is known classically; the oracle is a lookup. Tier A therefore demonstrates circuit correctness, not end-to-end speedup.**
- **Diffuser.** $U\cdot(2|1^k0^{L-k}\rangle\langle 1^k0^{L-k}| - I)\cdot U^\dagger = 2|D\rangle\langle D| - I$, an X-conjugated multi-controlled Z on all L qubits sandwiching the Dicke unitary; correctness holds for any unitary U with $U|1^k0^{L-k}\rangle = |D\rangle$.
- **Execution ladder.** (1) Aer ideal statevector and shot sampling; (2) Aer with a noise model from an IBM fake backend; (3) optional run on an IBM device through Qiskit Runtime if access exists. Transpile at optimisation level 3 and report depth and two-qubit gate count per Grover iteration; choose r to maximise measured success (noise shifts the optimum earlier). Report ideal vs noisy vs hardware success probabilities with shot-based confidence intervals. If hardware fails to amplify, report the noise-limited depth as a finding. Error mitigation: readout mitigation only, labelled; no post-selection on the known answer.

13.2 Tier B — exact reduced-space emulation

Setting. $H_k = \text{span}\{|x\rangle : |x| = k\}$, dim N. $|D\rangle = N^{-1/2}\,\Sigma|x\rangle$ is the uniform vector in the index basis. For a marked set $\mathcal{M} \subset \Omega_k$, $|\mathcal{M}| = M$:

- **Oracle** $O_f = I - 2\Pi_{\mathcal{M}}$, diagonal: $\psi_x \mapsto (-1)^{f(x)}\,\psi_x$. It maps H_k to itself.
- **Diffuser** $R_D = 2|D\rangle\langle D| - I$. In the index basis it acts as $\psi \mapsto 2\langle\psi\rangle\cdot\mathbf{1} - \psi$ (inversion about the mean amplitude).

- **Grover operator** $G = R_D O_f$, restricted to H_k, equals the standard N-dimensional Grover operator. Its dynamics live in span{|a⟩, |b⟩} with $\sin\theta = \sqrt{(M/N)}$, so $P_{success}(r) = \sin^2((2r+1)\theta)$.

Hence an N-dimensional classical amplitude simulation with "flip marked amplitudes; reflect about the mean" produces exactly the probabilities the ideal circuit produces. Equivalence is proven above and tested on toy instances against Aer (Test Q4).

Class-collapsed exact simulation. Starting from the uniform state, amplitudes remain uniform within the marked class (α_r) and within the unmarked class (β_r). With $\alpha_0 = \beta_0 = N^{-1/2}$, one iteration is $\alpha' = -\alpha_r$, $\beta' = \beta_r$, $\mu = (M\alpha' + (N-M)\beta')/N$, $\alpha_{r+1} = 2\mu - \alpha'$, $\beta_{r+1} = 2\mu - \beta'$. This is exact up to float64 error, uses O(1) memory per iteration, and sampling reduces to choosing the marked class with probability $M\alpha^2/(M\alpha^2 + (N-M)\beta^2)$ then a uniform member. The full-vector simulator is retained for toy cases as a cross-check (Test Q5). For N up to $\sim 10^7$ the full vector fits in memory (float32); beyond that the class-collapsed form is used.

> **Tier B is an exact classical emulation of the ideal quantum search dynamics. It is not a quantum runtime demonstration.** It validates the amplification dynamics, BBHT/Dürr–Høyer control logic, oracle-call statistics and success rates for a given predicate table. It does not validate realizability, noise or wall-clock behaviour.

13.3 Tier C — coherent fault-tolerant oracle (designed and estimated)

> **Status: DESIGNED + ESTIMATED. NOT SYNTHESIZED AT IEEE SCALE. NOT CURRENTLY FEASIBLE ON ANY DEVICE.** This section is an analytical cost model of a designed oracle. No circuit of this size has been synthesized or run.

Block	Content	Status
Search register	Option 1: L-qubit Dicke register. Option 2 (Tier C default): $\lceil \log_2 N \rceil$-qubit ranked index; diffuser reflects about the uniform superposition over [0,N)	DESIGNED
Branch mapping / Hamming constraint	Dicke: constraint intrinsic. Ranked: unranking r → sorted k-index list, ≈ O(kL log L) Toffolis via binomial QROM	DESIGNED, ESTIMATED (small)
Grid state	In-service register s (L bits), cumulative outage list K, per-round trip registers	DESIGNED
Fixed-point arithmetic	b-bit two's complement; schoolbook multipliers ≈ b^2 Toffolis; adders/comparators ≈ b	DESIGNED; primitive costs checked on A2
Data lookup (QROM)	Φ columns (≈ L Toffolis per lookup, output width L·b via CNOT fan-out), Φ_{KK} entries, base flows f^0, ratings, injections	DESIGNED, ESTIMATED
Island detection	Reversible label propagation over L branches, D ≈ graph-diameter iterations	DESIGNED, rough ESTIMATE
Island balance	Per-island generation/load sums, headroom redispatch, pro-rata scale	DESIGNED, ESTIMATED
PTDF injection update	After rebalancing, flows change by PTDF·ΔP (dense L×B product, ≈ $L \cdot B \cdot b^2$ Toffolis per round)	DESIGNED, ESTIMATED — dominant term
Rank-|K| flow update	$f' = f + \Phi_{:,K}(I - \Phi_{KK})^{-1} f_K$	DESIGNED, ESTIMATED
Overload comparison, trip update	Comparators vs ratings; s ← s ∧ ¬trip	DESIGNED
Cascade rounds	R rounds, R taken from measured cascade depth (E2)	DESIGNED
Severity and threshold	Served-load accumulation; compare against y or τ into a flag qubit	DESIGNED
Phase kickback	Flag qubit in |−⟩	Standard; IMPLEMENTED at A2 scale
Bennett uncomputation	Full reversal of all blocks	IMPLEMENTED at A2 scale; full scale ESTIMATED
IEEE-scale circuit	—	NOT SYNTHESIZED

13.4 Micro-oracle (A2)

Why it exists. Tiers A1 and B use classical lookups. A judge can fairly ask whether a coherent cascade oracle exists at all. A2 answers that at toy scale and supplies counted gate costs against which the analytical Tier C formulas are calibrated.

- **Instance.** A 4–5-bus toy grid with 5–7 branches, b ≈ 4–6-bit fixed-point DC flows, R = 2 cascade rounds, comparators against ratings, a served-load accumulator, a threshold comparison, phase kickback and Bennett uncomputation.

- **Verification.** (i) Exhaustive basis-state (permutation) simulation of the reversible circuit against the classical oracle on all 2^L inputs; (ii) all ancillas return to $|0\rangle$; (iii) the inverse circuit restores the input; (iv) Toffoli and qubit counts recorded per primitive.
- **Full Grover on A2.** Run on Aer only if the post-recycling qubit count is within statevector reach ($\lesssim 30$ qubits); otherwise A2 is validated through the equivalence chain (Test O1) and not run end-to-end.
- **Calibration.** κ_{cal} = (synthesized count)/(formula count) per primitive is applied to the resource model and reported with its range [MEASURED once E14 is run].

14. QAOA comparison

> **QAOA optimizes an explicit quadratic surrogate of S, never S itself.** Every figure label reads “surrogate objective”. QAOA has no proven speedup for this problem and is included as a near-term hybrid comparison and to expose the surrogate gap.

14.1 Surrogate

$$\hat{S}(x) = c_0 + \Sigma_i h_i x_i + \Sigma_{i<j} J_{ij} x_i x_j,\ x_i = (1 - Z_i)/2 \text{ (Ising mapping } H_C).$$

- **Training data.** $n_{train} \in \{100, 200, 400, 800\}$ uniformly sampled sets in Ω_k with true DC severity S; held-out test sets for R^2 and ρ.
- **Fitting.** Ridge regression; J sparsified to pairs that are neighbours, parallel or have large LODF coupling.
- **Identifiability.** With $|x| = k$ fixed, $\Sigma_i x_i = k$ and $\Sigma_j x_i x_j = (k-1)x_i$ on weight-k strings, so constant, linear and quadratic terms are partly degenerate. The effective rank of the design matrix is reported and ridge regularisation resolves the degeneracy.
- **k = 2: exact case.** S is a function of an unordered pair, so a full quadratic represents it exactly; the surrogate gap is zero and the experiment isolates QAOA’s optimizer performance. Exact fitting requires all C(L,2) pair evaluations, i.e. brute force, so QAOA has no oracle-call advantage by construction.
- **k = 3: interaction gap.** A quadratic cannot represent three-way interactions; the resulting gap is a finding, not a defect.

14.2 Ansatz and optimisation

- Dicke initial state; $p \in \{1,2,3,4\}$ layers of $e^{-i\gamma H_C}$ followed by a ring XY mixer (even/odd pair hopping) with parameter β; Hamming weight preserved exactly.
- COBYLA / SPSA from ≥ 10 starts with an interpolation warm start across p; 4096 shots.
- Simulation: exactly in the N-dimensional weight-k subspace (sparse mixer), cross-checked against Aer for $L \leq 12$; for $L > 24$ a candidate shortlist or MPS simulator is used and labelled.
- Report transpiled depth and two-qubit gate count per layer.

14.3 Mandatory metrics

#	Metric	Why mandatory
1	Surrogate R^2 and Spearman ρ, overall *and on the true top 1%*, on held-out sets	High overall R^2 can coexist with a useless tail
2	Approximation ratio against the **true** S_{max}, by verifying the top-T sampled strings with the true oracle	QAOA performance on the surrogate says nothing about the cascade by itself
3	Top-1 agreement and top-1% overlap, with enrichment over random	Measures retrieval quality
4	True-oracle calls = n_{train} + T	Fair accounting versus brute force (for IEEE-14 the fit alone costs more than enumerating N = 1,140)
5	Classical exact optimum of the same surrogate (enumerate N)	Trivial at L = 20; the meaningful comparator for QAOA

Error decomposition (key figure). (i) QAOA versus exact surrogate optimum = optimizer/depth error; (ii) surrogate optimum versus true optimum = model error. Experiments: E10a planted-quadratic sanity (QAOA recovers a known optimum), E10b k = 2 exact surrogate, E10c k = 3 surrogate gap.

15. Classical baselines and fair benchmark protocol

Classical methods are a first-class part of the study. Power systems are highly structured; sensitivities (PTDF/LODF), contingency ranking, mixed-integer optimization and local search routinely beat black-box enumeration. **Structured classical methods may outperform Grover on IEEE grids. That outcome is not hidden; it is a feature of the research question, which is about the conditions under which that stops being true.**

15.1 Baselines

Baseline	Search strategy	Objective	Evaluation accounting	Relevance
Brute force (batched DC)	Enumerate all N	S (DC)	N unique evaluations	Reference and the cost Grover is compared against. Feasible: IEEE-14 $k \le 4$, IEEE-30 $k \le 4$, IEEE-118 $k \le 4$
Random search	Uniform without replacement	S	Unique evaluations until S_{max} seen; $\approx N/(M+1)$	Baseline for threshold search
Hill climb + swap	Start from top-k N-1 outages; steepest ascent over swap neighbourhood; multi-start	S	Unique evaluations	**Strongest simple heuristic**; expected to win at $k = 3$ on IEEE-118
Beam search	Add branch maximizing S; width $w \in \{5, 20, 100\}$	S	Unique evaluations	Structure-exploiting
Simulated annealing	Swap moves preserving $\lvert x\rvert = k$	S	Unique evaluations	Standard metaheuristic
Genetic algorithm	Crossover/mutation preserving $\lvert x\rvert = k$	S	Unique evaluations	Standard metaheuristic
LODF/PTDF-guided pool	Rank branches by N-1 and pairwise LODF stress; exhaustively search the top-m pool, $m \in \{20, 40, 60\}$	S	Pool evaluations + ranking cost	Domain-informed screen
ML surrogate screen	Gradient-boosted trees on LODF/loading features; evaluate top-T predicted	S	$n_{train} + T$	Tests whether learned pruning erases N
DC interdiction MILP	Attacker–defender bilevel with DC redispatch (static, no cascade) via a MILP solver	Different severity definition	Solver time; chosen sets are re-evaluated under the cascade oracle	Strongest classical DC N-k technique; compared by ranking agreement and cascade-evaluated severity, **not** by call counts
Branch-and-bound [PROPOSED/stretch]	LODF-bounded pruning on DC	S	Nodes expanded	The classical outcome that would remove any crossover

15.2 Fair benchmark protocol

- Same oracle code (batched DC) for every method, with memoization; cost = **unique** sets evaluated, so duplicates do not inflate classical cost.
- Seeds ≥ 50 for stochastic classical methods; 200 for Dürr–Høyer/GAS emulation.
- Wall-clock on one core as well as the call-count metric.
- Success criterion: $S \ge S_{max} - 10^{-6}$ within budget; also report best S found, top-10/100 overlap with the reference, and calls to within 1% of S_{max}.
- Both *discovery calls* and *certified-stop calls* are reported (§12.3), because heuristics have no stopping guarantee.
- No baseline is deliberately weakened; hyper-parameters are tuned on a held-out grid/k pair and fixed before the comparison.
- Fitted scaling exponent α in unique-evaluations $\approx \kappa N^{\alpha}$ is reported for each baseline (E7); it feeds the structured crossover model of §16.4.

Pre-registered expectation H-C1. On IEEE grids, at least one structured baseline finds S_{max} in fewer unique evaluations than $c \cdot \sqrt{N}$ for most (grid, k) pairs, and exhaustive search is cheap enough that Grover's wall-clock is irrelevant. If measured, this is the headline result and is not softened. If amplitude amplification is lower in count and classical methods genuinely need $\gtrsim c\sqrt{N}$ evaluations, that is reported too.

Central framing. The motivation is not "quantum will beat classical search". It is: *quantum search may become relevant when the cascade predicate becomes sufficiently expensive and black-box, and classical*

structural pruning stops scaling.

16. Resource model, crossover and logical/physical estimates

> **Evidence status.** Everything in this section is **[DESIGN-STAGE]/[ESTIMATED]**. Numbers use placeholder parameters (R = 4, k_{eff} = 3k, t_c = 0.5 ms, c_q = 4) that are replaced by measured values from E1, E2, E6 and E14. They are not synthesized-circuit counts.

16.1 Model

Let N = C(L,k), M = 1 for the single worst set, c_q the empirical constant in $Q \approx c_q\sqrt{(N/M)}$, P the number of classical cores and P_q the number of quantum processors (Grover parallelizes only as $\sqrt{P_q}$; default 1).

$$Q = c_q\sqrt{(N/M)},\ G = 2E_f + 2C_D + C_{refl},$$
$$T_q = Q \cdot G \cdot t_T / \sqrt{P_q},\ T_c = N_c \cdot t_c / P.$$

E_f is the Toffoli count of one compute of f (the factor 2 covers compute and uncompute; Bennett uncomputation), C_D the Dicke-preparation cost per application (§11.3), $C_{refl} \approx$ L Toffolis for the multi-controlled Z (negligible), t_T the logical Toffoli time, t_c the classical time per evaluation, and N_c the number of classical evaluations (N for exhaustive search). Toffoli count is treated as serial depth, conservative for the quantum side; parallel arithmetic is a sensitivity case.

Exhaustive classical baseline: crossover

Setting $T_q = T_c$ with N_c = N and M = 1 gives $c_q\sqrt{N} \cdot G \cdot t_T/\sqrt{P_q} = N \cdot t_c/P$, hence

$$N^* = \left(c_q \cdot P \cdot G \cdot t_T / \left(t_c \cdot \sqrt{P_q} \right) \right)^2.$$

N* is quadratic in each of c_q, P, G and t_T: halving oracle cost lowers N* by 4×; a 10× faster logical clock lowers it by 100×; 1000 classical cores raise it by 10^6×. Because G depends on k through k_{eff}, a crossover read at k* must use G(k*); using G(k = 3) gives a lower bound on N*.

Structured classical baseline

If a classical method needs $N_c = \kappa N^{\alpha}$ evaluations, the crossover condition becomes

$$N^{\alpha - ½} = c_q \cdot G \cdot t_T \cdot P / \left(\kappa \cdot t_c \cdot \sqrt{P_q} \right).$$

If α ≤ ½ there is no asymptotic crossover. For α = ¾ and κ = 1 with the realistic placeholders below, the right-hand side is 4×10^6 and $N^* = (4\times10^6)^4 \approx 2.6\times10^{26}$. This is the central quantitative point behind framing B.

16.2 Oracle compute cost per cascade round (Toffolis)

b-bit fixed point; illustrative IEEE-118 values with b = 16, k_{eff} = m = 9 (placeholder 3k), B = 118 buses, L = 186 branches, graph diameter D ≈ 20 [DESIGN-STAGE].

Component	Formula	IEEE-118 example	In scenario
Island detection (label propagation)	$\approx L \cdot D \cdot c_{lab}$	$\approx 7\times10^4$ (rough)	Opt, Real, Pess
Rank-|K| flow update	$L \cdot m \cdot b^2$	4.29×10^5	Opt, Real
Small inverse $(I - \Phi_{KK})^{-1}$	$m^3b^2/3$	6.2×10^4	Opt, Real
Φ QROM lookups	$\approx m \cdot L$	1.7×10^3	Opt, Real, Pess
Injection update after rebalance (PTDF·ΔP)	$L \cdot B \cdot b^2$	**5.62×10^6**	Real, Pess
Comparators, trip update	$L \cdot b$	3.0×10^3	all
Served-load accumulation	$B \cdot b$	1.9×10^3	all
Full Gaussian elimination (pessimistic replacement for the two update rows)	$B^3b^2/3$	1.4×10^8	Pess
Per-round total		Opt $\approx 5.7\times10^5$; Real $\approx 6.2\times10^6$; Pess $\approx 1.5\times10^8$	
Once only: unranking/conversion, severity, threshold	$\approx O(kL \log L + b)$	$\approx 5\times10^3$	all

The injection update is the dominant term in the realistic scenario: whenever an island rebalances, bus injections change and the flow change is PTDF·ΔP, a dense L×B product. Because control flow is data-dependent, a fixed reversible circuit pays for it in every round. Omitting it, as in the optimistic scenario,

makes the oracle cost a **lower bound**.

16.3 Scenarios

	OPTIMISTIC	REALISTIC	PESSIMISTIC
Oracle blocks	Flow update + comparators + QROM + island detection; **no injection update**	All blocks above	Full reversible Gaussian elimination per round + all blocks
R, k_{eff}	Measured p50	Measured p95	Measured max (cap 30)
Fixed-point b	Minimum passing E12	16–20	24
Logical Toffoli time t_T	1 µs	10 µs	100 µs
Uncomputation	Measurement-based where valid	Full Bennett	Full Bennett, no savings
c_q	Measured mean	Measured mean	Measured p95
Qubits	Stream Φ columns	Hold them	Hold everything

Worked illustration [DESIGN-STAGE], IEEE-118, k = 3, N = 1,055,240, c_q = 4 (so Q ≈ 4.1×10^3 iterations), t_c = 0.5 ms DC on one core, R = 4 and k_{eff} = 9 as placeholders for all three scenarios:

Scenario	G (Toffolis / iteration)	t_T	T_q (k = 3)	T_c exhaustive DC, 1 core	N* (P = 1)	N* (P = 1000)
Optimistic	≈ 4.6×10^6	1 µs	≈ 5 h	≈ 8.8 min (528 s)	≈ 1.4×10^9 (≈ k = 5, lower bound)	≈ 1.4×10^{15}
Realistic	≈ 5.0×10^7	10 µs	≈ 24 d	≈ 8.8 min	≈ 1.6×10^{13} (between k = 7 and 8)	≈ 1.6×10^{19} (between k = 12 and 13)
Pessimistic	≈ 1.2×10^9	100 µs	≈ 16 yr	≈ 8.8 min	≈ 9×10^{17}	≈ 9×10^{23}

Reading the table: at k = 3 on IEEE-118, classical exhaustive search wins in every scenario, by roughly 1.5 (optimistic) to 6 (pessimistic) orders of magnitude. Even the quantum-favourable optimistic scenario — which ignores injection updates — only approaches a crossover near k ≈ 5, and that against a single classical core running exhaustive search, and with G evaluated at k = 3. Against 1000 cores, or against a classical method with α < 1, the crossover moves out by orders of magnitude.

16.4 Logical and physical qubits

- **Logical qubits [DESIGN-STAGE], realistic scenario.** In-service register and retained per-round flow/trip registers R(Lb + L) ≈ 1.3×10^4; held Φ columns m·L·b ≈ 2.7×10^4; arithmetic workspace. Total of order 4×10^4. Pessimistic design holding the dense matrix needs ≈ B^2·b ≈ 2×10^5 logical qubits for the matrix alone.
- **Physical qubits [ESTIMATED, bound quality].** Total Toffoli volume Q·G ≈ 2×10^{11} pushes the surface-code distance to roughly d ≈ 25–31 at physical/threshold error ratio near 0.1, i.e. ≈ 10^3 physical qubits per logical qubit, plus magic-state factories and routing: **10^7–10^8 physical qubits**.
- **Caveats.** Factory footprint, routing, T-count versus Toffoli accounting, code-distance dependence on total volume and the logical-clock assumption are not modelled in detail. These are bounds, not synthesized figures.

16.5 What can be estimated and what remains a bound

Estimated from experiments	Remain bounds / assumptions
t_c (E1), c_q (E6), R and k_{eff} (E2), minimum b (E12), C_D calibration and per-primitive Toffoli counts (E14), classical exponent α (E7)	t_T, factory overhead, true oracle size at IEEE scale, island-logic cost, error-correction overhead

17. Sensitivity analysis

Factor	Range	Why it matters	Metrics	Placement
Headroom h	0.10–0.50	Defines how fragile every line is	Top-10/100 Jaccard, Spearman, #high-severity, high-severity fraction	Main
Stress λ	1.0–1.3	Moves the grid toward its limit	same	Main
Trip factor τ_{trip}	1.0, 1.1, 1.2, 1.25	Models relay pickup margin	same	Main
Rating family	proportional to flow; nominal MVA by voltage class; case-file ratings; ±10% perturbation (50 draws)	Tests whether the rating rule creates the vulnerabilities	Top-10 stability	Main
Trip policy	simultaneous / sequential	Determinism and cascade depth	Ranking, rounds	Main (short)
Redispatch variant	with / without upward ramp	Optimism of balance rule	S shift, ranking	Appendix
Max rounds	5, 10, 30	Truncation vs EXHAUSTED fraction	EXHAUSTED fraction, S shift	Appendix
AC tolerance	10^{-6}, 10^{-8}, 10^{-10}	Convergence classification	S interval width	Appendix
Load-shedding ladder	1%, 5%, 10%, total loss	Weight of the modelling abstraction	Interval [S_{ladder}, S_{total}], top-10	Main
DC approximation	E3	Fidelity of the search predicate	Recall, ρ, false negatives	Main
Fixed-point b	12, 16, 20, 24, 32	Precision a coherent oracle needs	see §17.1	Main
Effective update size k_{eff}	2k, 3k, p95, max	Truncation of the oracle; resource cost	Truncation error, Toffolis	Main
Toffoli time t_T	1–100 µs	Quadratic in N*	N*, T_q	Main (tornado)
c_q	measured; p95	Quadratic in N*	N*	Main (tornado)
t_c	0.05–10 ms	Quadratic in N* (inverse)	N*	Main (tornado)
P	$1–10^5$	Quadratic in N*	N*	Main (tornado)

17.1 Fixed-point precision protocol

Emulate Q-format arithmetic in the DC cascade (4 integer bits, b−4 fractional, per-unit on 100 MVA, round to nearest) and truncate to R rounds. For b ∈ {12, 16, 20, 24, 32} report: max and mean flow error; severity error; high-severity label agreement; top-10 and top-100 overlap; Spearman ρ; the **borderline fraction** (overload comparisons within quantization error of the threshold, which predicts label flips); the distribution of the condition number of $(I - \Phi_{KK})$ (error amplification); and the resulting Toffoli count. The smallest b that preserves labels and top-10 feeds the resource model. Tolerance sweeps for AC and the full resource re-estimate across all five widths go to the appendix; the main text carries the minimum passing b.

18. Experiment plan E0–E16

> All “expected” entries are **hypotheses to be tested**, not results. No experiment has been run at the time of writing. Priority: M = must, S = should, N = nice to have.

ID	Name	Purpose	Method / metrics	Expected (hypothesis)	Failure interpretation	Pri
E0	Simulator validation	Is the twin correct?	Toy 3/4/5-bus grids; pandapower base and N-1; hand-solved cascades. Metrics: flow error, conservation residual, cascade sequence match.	All match to tolerance	Do not proceed; every later result is void	M
E1	Runtime profile	Measure t_c	1000 random + 1000 high-severity sets, DC and AC, IEEE-14/30/118. Mean, p95, split by class (islanding, long cascade).	DC $\lesssim$ 0.5 ms, AC $\lesssim$ 30 ms (targets)	Replace placeholders; widen crossover bands	M
E2	Cascade phenomenology	RQ1; supplies R, k_{eff}	Full DC tables: IEEE-14 k $\leq$ 4; IEEE-30 k $\leq$ 4; IEEE-118 k $\leq$ 4. S distribution, #high-severity, rounds, trips (p50/p95/max), C(x), non-monotonicity fraction.	Heavy-tailed S; most severe sets cascade-driven or near-bridge	If nearly all severity is S_0 (isolation), the cascade adds little and the problem is trivial	M
E3	DC–AC agreement	RQ2	E3a/b/c of §10.1.	Good rank agreement with non-zero false negatives	Poor agreement: fidelity ladder becomes the finding; DC-only claims restricted	M
E4	Oracle and circuit equivalence	RQ4	Dicke fidelity (L $\leq$ 10); reflection operator; phase-oracle diagonal; Aer vs Tier B probabilities; micro-oracle exhaustive check.	Fidelity $\geq 1 - 10^{-10}$; probabilities agree to 10^{-9}	Circuit bug; Tier A claims withdrawn	M
E5	Grover dynamics, ideal and noisy	Noise behaviour	Toy N = 15–56; r sweep on Aer ideal and noise model; P(marked) vs $\sin^2$; depth; 2-qubit gates; optimal r under noise.	Ideal matches theory; noise lowers the peak and moves r^* earlier	Failure to amplify is reported as a noise-limited-depth result	S (ideal: M)
E6	GAS/Dürr–Høyer scaling	RQ4	Tier B on E2 tables, 200 seeds, $y_0 = 0$ and heuristic y_0. Mean/p95 calls, success rate, log-log slope, fitted c with CI, discovery vs certified-stop calls.	Slope $\approx$ 0.5; c below 22.5	Slope far from 0.5: implementation or landscape issue	M
E7	Classical comparison	RQ5	All §15 baselines. Unique evaluations to S_{max}, success at budget, wall-clock, top-10 overlap, fitted α.	Structured methods find S_{max} in $\ll c\sqrt{N}$ evaluations	If quantum count is lower and classical needs $\gtrsim c\sqrt{N}$, report it; baselines are not tuned down	M
E8	Threshold regime map	Problem 3	f_τ over a τ sweep: M/N versus BBHT calls versus random-sample calls.	Random sampling wins when M/N is large	Shows threshold search can be classically trivial	S
E9	Top-T cost	Problem 5	T $\in$ {1, 5, 10, 50}: calls versus $2\sqrt{(NT)}$.	Tracks $\sqrt{(NT)}$	Check exclusion-list logic	N
E10	QAOA (a, b, c)	RQ7	§14: planted-quadratic; k = 2 exact surrogate; k = 3 gap. Five mandatory metrics.	(a) recovers planted optimum; (b) near-optimal; (c) a gap	(a) fails: QAOA bug; (c) low tail R^2: surrogate invalid	S
E11	Rating/trip/ladder robustness	RQ3	§17 main grid. Jaccard, Spearman, high-severity counts, cascade-driven fraction.	Moderate stability	Unstable top-10: findings are artefacts of the rating rule; reported as such	M
E12	Fixed-point and truncation	Oracle precision	§17.1 protocol.	b = 16 passes; b = 12 shows label flips	b $\geq$ 20 needed: resource model rises accordingly	S
E13	Hardware	A1 on device	A1 toy on an IBM device.	Likely noise-dominated	Reported as a finding	N [OPTIONAL]
E14	Micro-oracle synthesis and calibration	Cost-model calibration	A2: measured Toffoli and qubit counts versus formulas; κ_{cal}.	κ_{cal} within $\approx$ 0.5–2	$\kappa \gg 1$: oracle cost underestimated	S (high value)
E15	Resource and crossover sensitivity	RQ6	Scenarios $\times$ (t_T, k_{eff}, R, b, P, α); tornado and 2-D heat maps.	N^* far above IEEE sizes	Crossover close: re-examine the classical exponent α	M

ID	Name	Purpose	Method / metrics	Expected (hypothesis)	Failure interpretation	Pri
E16	Falsification summary	All	Apply pre-registered criteria to E3, E7, E11.	—	—	M

18.1 Pre-registered falsification criteria

- **F1.** A structured classical method finds the AC-verified S_{max} in $\ll \sqrt{N}$ evaluations at all tested sizes: the quantum-search motivation is then empirically empty on these grids.
- **F2.** DC recall at τ falls below the threshold of §10.1: the DC predicate is the wrong oracle.
- **F3.** Top-10 Jaccard across rating families falls below ≈ 0.3: the findings are artefacts of the rating rule.

18.2 Result tables to be filled from runs

Grid, k	N	M (S ≥ τ)	Best S (DC)	Best S (AC)	C(x*)	DC–AC top-100 overlap	Ladder fraction
IEEE-14, 3	1,140	[to be filled]	[ ]	[ ]	[ ]	[ ]	[ ]
IEEE-30, 3	10,660	[to be filled]	[ ]	[ ]	[ ]	[ ]	[ ]
IEEE-118, 3	1,055,240	[to be filled]	[ ]	[ ]	[ ]	[ ]	[ ]

Grid, k	Method	Discovery calls (mean / p95)	Certified-stop calls	Success @ budget	Wall-clock	Notes
IEEE-14, 3	Dürr–Høyer (Tier B)	[ ]	[ ]	[ ]	[ ]	[EMULATED]
IEEE-14, 3	Hill climb / beam / SA / GA / random / LODF pool / ML / MILP	[ ]	n/a	[ ]	[ ]	[MEASURED]

19. Verification and testing

Tests are organised as unit (U), integration (I), quantum (Q), oracle-equivalence (O) and regression/reproducibility (R) suites. CI runs U, Q, I and R on IEEE-14 data in minutes; heavy tables run under `make truth`. Status of all tests at the time of writing: [PROPOSED].

ID	Test	What can fail?	What does passing prove?
U1	Hand-solved 3/5-bus DC flows (10^{-8})	Susceptance sign, slack handling, incidence orientation	DC solver implements $B'\theta = P$
U2	Hand/pandapower AC 3-bus	Y-bus, PV/PQ handling	AC solver correct on a solvable case
U3	PTDF/LODF vs full re-solve, all N-1 + random K (10^{-9})	Matrix algebra, ordering	Sensitivity matrices are right
U4	Woodbury vs full re-solve, K up to size 6	Compensation formula, sign	Fast DC path equals exact path
U5	Island detection $\Leftrightarrow \Phi_{KK}$ singular, random K vs NetworkX	Connectivity logic, tolerances	Islanding is detected exactly; no silent division
U6	Island balance conservation	Redispatch/shedding bugs	Mass balance holds in every island
U7	AC vs pandapower base and N-1 (10^{-6} p.u.)	Convention mismatch	AC engine reproduces a reference solver
U8	Rating rule: base secure, floor, N-1 secure per engine	Calibration bug, MW/MVA mismatch	The grid is genuinely secure before searching
U9	Hand-solvable cascades (5-bus two-round; 4-bus island; parallel line; no-generator island)	Trip logic, round semantics	Cascade engine matches known sequences
U10	Invariants: $S \in [0,1]$, $S(\varnothing) = 0$, order-invariance of x, tie rule	Indexing/ordering bugs	Oracle is a well-defined function of the set
U11	AC non-convergence ladder: constructed infeasible case, determinism, flagging	Ladder nondeterminism	Fallback is deterministic and reported
U12	Fixed-point emulation vs float; error $\propto 2^{-b}$	Overflow, rounding bug	Precision study is trustworthy
I1	Batched ≡ scalar evaluation; chunking and cache-key invariance	Vectorisation bugs, stale cache	Tables are correct and reproducible
I2	Repeat runs, thread counts, seeds	Race conditions, uncontrolled RNG	Determinism to 10^{-9} (not bit-identity)
I3	Oracle-contract conformance for every oracle	Impurity, hidden state	Search code is oracle-agnostic
Q1	Dicke circuit vs analytic, $L \le 10$, all k ($F \ge 1 - 10^{-10}$); support only on weight k	Gate-order errors	Dicke preparation is exact
Q2	Reflection operator $= 2\|D\rangle\langle D\| - I$ on the weight-k subspace ($L \le 8$)	Wrong sign or phase	Diffuser is the correct reflection
Q3	Phase-oracle circuit diagonal = mask	Wrong control polarity	Circuit oracle equals the classical mask
Q4	Aer Grover probabilities = Tier B (10^{-9}), all r	Convention mismatch	The emulation is equivalent to the circuit
Q5	Class-collapsed ≡ full-vector ≡ $\sin^2$ formula	Recursion error	Tier B is exact
Q6	Micro-oracle: exhaustive basis simulation = classical oracle; ancillas clean; inverse restores	Garbage qubits, wrong uncompute	The coherent oracle idea works at toy scale
Q7	Zero-noise sim = ideal; noise degrades monotonically	Noise-model wiring	Noisy results are interpretable
Q8	Dürr–Høyer/BBHT on synthetic distinct and tied tables: mean calls ≤ bound, success $\ge 1 - \delta$, termination	Infinite loops on ties, bad stopping	Search logic is correct independent of the grid
O1	Oracle equivalence chain: DC-batched ≡ DC-scalar ≡ Tier A mask ≡ Tier B mask ≡ micro-oracle	Any link breaks	All tiers evaluate the same predicate
R1	Cost model vs three hand-computed rows; N* vs numerical root of $T_q = T_c$; $\alpha = 1$ limit; A2 calibration	Algebra, units	Resource model is arithmetically correct (not that its assumptions are)
R2	Golden-file regression on IEEE-14 (tolerance)	Silent behaviour change	Refactors do not alter results
R3	Figure provenance: every figure regenerates from `make figs` with config hash	Hand-edited numbers	Every plotted number is reproducible
R4	Claims lint: scans deck, README and PDF for banned phrases (“ground truth”, “quantum advantage”, “blackout prediction”, “validated” without linked experiment)	Overclaiming drift	Language matches the evidence ledger

20. Software and API architecture

```
doomsday/
├── gridtwin/        # L0–L2: no quantum imports
│   ├── loader.py  model.py (Grid dataclass)  ratings.py  scaling.py
│   ├── dc.py (PTDF/LODF/Woodbury)  ac.py (NR, ladder)  islanding.py
│   ├── cascade.py (one engine, mode = dc|ac)  severity.py (S, S0, C, A, flags)
│   └── contingency.py (Ω_k enumeration, ranking/unranking)
├── oracle/          # L3: the ONLY interface search code sees
│   ├── interface.py (severity, severity_batch, fidelity, cost_counter)
│   ├── dc_oracle.py  ac_oracle.py  table_oracle.py (cached DC/AC tables)
│   └── cache.py     (key: grid hash + config hash + code version)
├── qsearch/         # depends ONLY on oracle.interface (or a mask/callable)
│   ├── dicke.py  circuits.py (Tier A1)  microoracle.py (A2)
│   ├── emulation.py (Tier B, class-collapsed)  grover.py  bbht.py  dh_gas.py
│   └── counting.py (analytic/table-derived)  qaoa.py  surrogate.py
├── classical/       # depends ONLY on oracle.interface
│   └── brute.py random.py hillclimb.py beam.py sa.py ga.py lodf_pool.py ml_screen.py milp.py
├── resource/        # depends on measured JSON + (L, B, k) only
│   └── components.py scenarios.py crossover.py qubits.py calibration.py
├── experiments/     # E0..E16 scripts -> results/*.json + figures
├── api/             # FastAPI: /cascade /search /crossover (precomputed-first)
└── app/             # static D3 front end + precomputed JSON frames
tests/  configs/  docs/  Makefile  pyproject.toml
```

```
                gridtwin ──► oracle.interface ◄── qsearch
                                    ▲  ▲
                                    │  └──── classical
       experiments ─────────────────┘
       resource  ◄── measured results (JSON) only; never imports gridtwin
       api / app ◄── result JSON; optional live calls to oracle for IEEE-14/30
```

20.1 Module boundaries

- **qsearch and classical depend only on** `oracle.interface`: swapping the DC oracle, AC oracle, cached table or micro-oracle never touches search code. An import-linter in CI enforces this.
- **resource is decoupled**: it consumes measured JSON (t_c, c_q, R, k_{eff}, κ_{cal}) and (L, B, k), and never imports gridtwin.
- **Results are reproducible**: dependencies pinned, versions written into each results file, every random source seeded from per-experiment seed lists, truth tables cached by (grid hash, config hash, code version).
- **Figures are generated from configs**: every figure and table in the deck comes from a script in `experiments/` (Test R3); no hand-made numbers.

```
class Oracle(Protocol):
    fidelity: Literal["dc", "ac"]
    def severity(self, outage: Sequence[int]) -> float: ...            # S(x) in [0,1]
    def severity_batch(self, outages: np.ndarray) -> np.ndarray: ...   # vectorised, shape (n,k)
    def is_high_severity(self, outage, tau: float) -> bool: ...
    cost_counter: CallCounter                                           # unique evaluations

def run_cascade(grid, outage, cfg) -> CascadeResult:   # S, S0, C, rounds, trips, islands, status, flags, frames
def classical_search(oracle, L, k, method, budget, seed) -> SearchResult
def dh_max(table_or_oracle, N, y0, budget, seed) -> SearchResult
def build_grover_circuit(L, k, marked, r) -> QuantumCircuit
def qaoa_search(surrogate, L, k, p, optimizer, shots, seed) -> SearchResult
def oracle_toffoli(L, B, k, scenario, b, R) -> dict            # per-component breakdown
def crossover_N(G, t_T, t_c, P, c_q, Pq=1, kappa=1, alpha=1) -> float
```

20.2 Configuration

```
grid: ieee118
k: 3
rating: {family: proportional, headroom_h: 0.25, floor_frac: 0.05}
stress_lambda: 1.10
cascade: {trip_factor: 1.0, trip_policy: all_overloaded, max_rounds: 30, mode_default: dc,
          ladder_step: 0.05, redispatch: up_and_down}
severity: {tau: 0.20, delta_cascade: 0.10}
quantum: {encoding: binary_dicke, bbht_lambda: 1.2, seeds: 200, patience: 3, y0: 0.0}
resource: {b: 16, R: 4, k_eff: 3k, t_toffoli_us: 10, c_q: 4.0, t_classical_ms: {dc: 0.5, ac: 10}, cores: [1, 1000]}
```

20.3 API architecture

Endpoint	Function	Data source
`GET /grids, /grid/{id}`	Grid topology, layout, base loading	Precomputed JSON
`POST /cascade`	Run or fetch a cascade for an outage set; returns frames and metrics (S, S_0, C, status, ladder interval)	Precomputed table lookup; live DC/AC for IEEE-14/30

Endpoint	Function	Data source
`POST /search`	Return Tier B emulation logs and classical baseline traces for a (grid, k, seed)	Precomputed experiment logs; optional live emulation
`GET /ranking`	Top-T sets with DC and AC values and cascade-driven flags	Result tables
`POST /crossover`	Evaluate the resource model for slider inputs	Cost model (JS port verified against Python, Test R1)

The default demonstration mode is static: a front end reading precomputed JSON, so it cannot fail on venue networks. Live recomputation is optional and limited to IEEE-14/30.

21. Web demonstration

The demo must communicate the concept to a judge who knows nothing about quantum computing, and it must never let a viewer mistake emulation or modelling for measurement. Every panel carries a permanent four-chip legend: **[MEASURED] [EMULATED] [MODELLED] [PROPOSED]**.

Step	View	Content
1	Select grid	IEEE-14/30/118 on a stored force-directed layout; branch colour = loading (green ≤ 0.8, amber 0.8–1.0, red > 1.0)
2	Select k	Slider with live counter “C(186,3) = 1,055,240 possible outage sets” and the combinatorial-wall chart with the current point highlighted
3	Example outage sets	Three cards: benign; isolation-only; cascade-driven
4	Run cascade	Click a set; press play
5	Visualise propagation	Round scrubber; trips flash and grey out; de-energised buses dark; side plot of load served per round
6	Severity	S, S_0, C(x), cascade-driven badge; DC/AC toggle showing the interval [S_{ladder}, S_{total}] and any ladder flag
7	Worst set	Ranked top-10 table (S, C, rounds, fidelity); select a row to replay
8	Run quantum search (emulation)	Bars of Grover amplitudes for the top-20 states per iteration with marked states highlighted; success-probability curve with the $\sin^2((2r+1)\theta)$ overlay; chip [EMULATED]
9	Compare classical search	Live race plot: unique oracle calls versus best S found for Dürr–Høyer (emulated), random, hill climb, SA and brute force
10	Oracle calls	Footer: “unique oracle evaluations: …; table size: …; fidelity: DC”
11	Resource crossover	Sliders for t_T, P, oracle scenario (Opt/Real/Pess) and α; plot of T_q versus T_c with N* and a marker for the current (grid, k); annotation “IEEE grids sit left of every crossover” if the measured data say so; chip [MODELLED]
12	Demonstrated vs theoretical	Permanent legend and a final panel listing exactly what is measured, emulated, modelled and proposed

Layperson mode. One sentence — “Of 1,055,240 ways to lose three lines, these three cause the largest load loss in our model” — and a before/after city-lights toggle. The layperson text never says “blackout prediction”.

Implementation notes. IEEE cases have no geographic coordinates; a Kamada–Kawai layout is generated with NetworkX, hand-adjusted for 14 and 30 buses, and stored as `layout.json`. Static D3 (or Cytoscape.js) with precomputed JSON; the crossover explorer is a JavaScript port of the Python cost model verified by Test R1.

21.1 Cascade frame schema

```
{ "grid":"ieee118", "outage":[...k indices...], "mode":"dc", "policy":"all_overloaded",
  "metrics":{"S":0.0,"S0":0.0,"C":0.0,"rounds":0,"status":"STABLE","ladder_step":null,"S_total":null},
  "frames":[ {"round":0,"tripped":[...],"load_served":0.0,"islands":1,
              "loading":{"<branch>":0.0},"energized_buses":[...]} ] }   # all numbers from runs
```

22. Claims and evidence policy

Each claim is classed SAFE, CONDITIONALLY SAFE or UNSUPPORTED. An automated claims lint (Test R4) scans the PDF, README and deck for banned phrases.

Claim	Status	Safe wording
“Quantum advantage”	UNSUPPORTED	We estimate under which conditions a √N query reduction could become a wall-clock advantage; none is demonstrated.
“Grover finds the worst blackout faster”	UNSUPPORTED	Amplitude amplification needs ≈ c√N oracle queries in emulation, versus N for exhaustive search.
“√C(L,k) speedup”	CONDITIONALLY SAFE	“√C(L,k) query complexity; no wall-clock claim.”
“Blackout prediction”	UNSUPPORTED	“Ranking of high-severity outage sets under a quasi-steady-state cascade model.”
“Ground truth”	UNSUPPORTED for the grid	“Exhaustive DC reference table”; “AC verification”. Exact reference is allowed only when scoped: “exact for the DC predicate”.
“Realistic cascade”	UNSUPPORTED	“Quasi-steady-state overload cascade (no dynamics, protection timing or operator action).”
“Real-world deployment”	UNSUPPORTED	“The oracle interface is designed to accept other network models; operational use would require utility data and validation.”
“Validated”	CONDITIONALLY SAFE	Only “cross-checked against pandapower and hand-solved cases (U1–U9); DC–AC agreement measured in E3”, and only after those are run.
“Optimized”	CONDITIONALLY SAFE	“Searched for the highest-severity set.”
“Accurate”	UNSUPPORTED	Report the measured error numbers.
“Solves”	UNSUPPORTED	“Searches” / “identifies … under this model”.
“Demonstrates”	CONDITIONALLY SAFE	Only for implemented items; “emulates” for Tier B; “estimates” for Tier C.
“Practical”	UNSUPPORTED	“Not practical on current hardware; see crossover conditions.”
“Circuits executed on Aer (toy)”	SAFE once run	Verbatim, with N and noise model stated.
“Tier B emulates idealised Grover exactly”	SAFE	Verbatim.
“Classical wins on IEEE grids”	CONDITIONALLY SAFE	Safe only after E7 measures it; until then “expected”.

22.1 Implementation-status ledger

Component	Category
DC/AC cascade, truth tables, classical baselines, QAOA statevector, web app	IMPLEMENTED when built; each flagged live in the README
Tier A1 circuits on Aer (ideal, noisy)	IMPLEMENTED; executed on a simulator
IBM hardware run	OPTIONAL; report whatever happens
Tier B class-collapsed Grover/Dürr–Høyer at IEEE scale	EMULATED (exact classical emulation)
A2 micro-oracle	IMPLEMENTED at toy scale; verified by exhaustive basis-state simulation
Tier C oracle at IEEE scale	DESIGNED + ESTIMATED; never synthesized
Fault-tolerant resources and crossover	ESTIMATED
Real-utility operation	NOT CURRENTLY FEASIBLE; future validation target

23. Limitations and threats to validity

23.1 Limitations

- **No transient dynamics, frequency response, generator or governor dynamics.** Island balance is a static abstraction.
- **No protection timing, hidden failures, relay miscoordination or operator intervention.** The trip rule is a fixed overload multiple, not a relay model.
- **Synthetic ratings.** Findings characterise the model; they are not utility data.
- **DC approximation.** DC omits voltages, reactive flow and losses and can miss voltage-driven failures; its fidelity is measured (E3), not assumed.
- **AC is static verification only.** Non-convergence is flagged and bounded, not interpreted as proven collapse; reactive limits are off by default.
- **Coherent IEEE-scale oracle is not currently feasible** and has not been synthesized; Tiers A1 and B oracles are classical lookups by design.
- **Current quantum hardware** can run only toy instances (N = 15–56), likely noise-limited.
- **Resource model uncertainty.** Island logic, factory overhead, routing and the logical clock are bounds; R and k_{eff} are placeholders until E2.
- **Classical structure.** Structured classical methods are expected to outperform unstructured search on these grids.
- **IEEE benchmark limitations.** Small, public test systems; behaviour at thousands of branches is not measured.
- **No real utility data** (often access-restricted) and no real-grid validation.

23.2 Threats to validity

Threat	Mitigation
Synthetic ratings and stress define fragility	E11 sensitivity over four rating families and perturbations; Jaccard/Spearman stability
Deterministic trip rule is not a relay model	Pickup-margin sweep; statement of scope
DC–AC difference	E3a–c with false-negative analysis; AC verification of every reported set
AC ladder is an abstraction	Interval reporting; fraction reported; ladder sensitivity
Fitted constant c depends on the table	Reported with CI per (grid, k); both y_0 regimes
Baseline weakness	Fair protocol, tuning on held-out pair, strongest baselines included
Placeholder resource parameters	Replaced by measurements; scenario bands; tornado analysis
Emulation mistaken for quantum performance	Status chips on every panel; claims lint

23.3 Why the project remains valuable

The value does not depend on a quantum win. The project supplies a reproducible, oracle-agnostic benchmark for N-k cascading search; a measured account of how far a DC screen can be trusted against AC; a verified circuit-level and emulated implementation of constrained amplitude amplification; a toy-scale coherent oracle that grounds the cost model; and a quantitative statement of the conditions under which a fault-tolerant quantum oracle could compete with structure-exploiting classical methods. Each is useful whichever way the comparison comes out.

24. Judge questions and answers

Question	Best 20-second answer	Evidence	Weakness to admit / answer to avoid
Where is your quantum advantage?	There is none on these grids and we measured that. We quantified the oracle size, clock speed and classical baseline an advantage would need.	E6, E7, E15	No end-to-end speedup exists even in the optimistic scenario at IEEE scale. Avoid: “Grover gives √N so it’s faster.”
Isn’t your oracle a lookup table?	In A1 and B, deliberately. A2 is a synthesized reversible cascade oracle verified exhaustively; Tier C is an estimate.	Q4, Q6, E14	IEEE-scale oracle not synthesized. Avoid: “Our oracle is quantum.”
Why not brute force?	We do; it is the reference up to IEEE-118 k = 4. At k ≥ 5 it is ≈ 200 days at 10 ms per evaluation on one core.	E1, E2	On IEEE grids brute force is practical. Avoid: “Brute force is impossible.”
Why not MILP / interdiction?	It is the strongest classical method and a baseline. It answers a related static question, so we compare ranking agreement and evaluate its sets under the cascade.	E7	May beat us on DC-like predicates. Avoid: “MILP can’t handle cascades.”
Why not LODF, ML or GNNs?	They are baselines and will probably win here; learned pruning is exactly what would erase the crossover.	E7, E15	No GNN unless time allows. Avoid: “ML can’t be exact.”
Why Grover on a structured problem?	To ask when a query-optimal black-box method matters; structure is what classical methods exploit, so we measure it.	E7, E15	The grid is not unstructured. Avoid: “Power grids are unstructured.”
Is DC valid?	As a screening model with measured fidelity, not as truth.	E3a–c	DC can miss voltage-driven failures. Avoid: “DC is accurate.”
What happens when AC does not converge?	We flag static infeasibility, apply a documented curtailment ladder as a numerical proxy, and report results with and without it.	U11, E11	The ladder is an abstraction. Avoid: “That’s voltage collapse.”
Why do synthetic ratings represent reality?	They do not claim to; they are N-1-secure by construction and we test whether conclusions survive rating-rule changes.	E11	Results characterise the model. Avoid: “Realistic ratings.”
Why trust the cascade model?	A standard quasi-steady-state overload cascade, cross-checked against pandapower and hand cases, used for screening, not prediction.	E0, U1–U9	No dynamics or relay timing. Avoid: “It predicts blackouts.”
Where are frequency and voltage dynamics?	Out of scope and stated; static balance and AC non-convergence are weak proxies.	Limitations	Avoid: “We model voltage collapse.”
Where is protection modelling?	A fixed overload trip rule with a pickup-margin sweep; no relay timing or hidden failures.	E11	Avoid: “Our trip rule models relays.”
Why exactly k? Why not ≤ k?	Exactly-k fixes Hamming weight so the Dicke state is exact; S is not monotone so ≤ k differs; we sweep k.	E2	Convenience of the encoding. Avoid: “Larger k is always worse.”
Doesn’t Dicke preparation destroy the advantage?	No: O(kL) gates against ≈ 10^7 Toffolis per oracle call; it is an explicit term in the model.	Q1, E14	Rotation synthesis cost is a calibrated parameter. Avoid: “Dicke prep is free.”
Actual qubit count and depth?	Toy: measured after transpilation. IEEE-118: ≈ 4×10^4 logical (realistic), 10^7–10^8 physical, as bounds.	E5, E14, E15	Not synthesized. Avoid: quoting 10^7 as precise.
Can today’s hardware run this?	Only the toy, and likely noise-limited; IEEE scale needs fault tolerance.	E5, E13	Hardware may fail to amplify. Avoid: “We ran the grid on IBM hardware.”
Why QAOA, and what does the surrogate mean?	A near-term hybrid comparison on a fitted quadratic proxy; we separate optimizer error from model error.	E10	Fit cost exceeds brute force at IEEE-14. Avoid: “QAOA finds the worst set.”
Does the top-10 change under small parameter changes?	We measure it with Jaccard/Spearman across h, λ, trip factor and rating families.	E11	If unstable, that is a result. Avoid: “It’s robust” before running E11.
What if a classical heuristic beats Grover 1000×?	We expect that on IEEE grids and report it.	E7, α analysis	Then the conditional crossover analysis is the remaining claim. Avoid: hiding or re-tuning baselines.
What is implemented vs simulated vs theoretical?	The status ledger and on-screen chips: MEASURED, EMULATED, MODELLED, PROPOSED.	§22.1	Keep the ledger honest. Avoid: describing the design as built.
What would falsify your hypothesis?	Pre-registered: structured classical wins at all sizes; DC recall too low; top-10 unstable across rating families.	E16	Any may happen. Avoid: “Nothing could.”

Question	Best 20-second answer	Evidence	Weakness to admit / answer to avoid
What about fixed-point precision?	We sweep 12–32 bits and report label agreement and top-10 overlap; the smallest passing b feeds the model.	E12	16 bits was an assumption until E12. Avoid: “16 bits is enough.”
Is your GAS constant empirical or proven?	Proven bound $22.5\sqrt{N} + 1.4 \lg^2 N$; ours is a fitted c with a confidence interval, landscape-dependent.	E6	c depends on the table. Avoid: treating c as universal.
Why doesn’t √N queries mean a faster computer?	Each query is a long reversible circuit on a slow logical clock: $\approx 5\times10^7$ Toffolis per iteration versus a ≈ 0.5 ms classical evaluation (placeholders).	§16, E15	Depends on placeholders until measured. Avoid: equating queries with time.
What about classical parallelism?	Classical search parallelises almost perfectly, Grover only as $\sqrt{P_q}$; both are sliders.	E15	Avoid: “Grover also parallelises perfectly.”
How do you define “worst” with ties?	S_{max} with a 10^{-9} band; success means finding any set at S_{max}.	U10, Q8	Multiple worst sets are common. Avoid: “The unique worst set.”
Is the DC-worst set also the AC-worst?	We measure it: top-10 overlap and rank correlation from the full AC table at $k \leq 3$.	E3a	Not guaranteed. Avoid: assuming it.
What about real utility deployment?	The oracle interface accepts other models, but real use needs utility data, real ratings and dynamic validation.	Limitations	Not claimed. Avoid: “Deploy as is.”
Why does this matter?	N-k screening is a real planning problem; the output shows what a quantum oracle must achieve before it matters.	§2, E15	Value is mostly methodological. Avoid: “It will replace operator tools.”
What did you build and what did you reuse?	The ledger names every module; each team member can derive and rerun theirs. Reused: case data, pandapower, Qiskit.	Repo, R2–R4	Avoid: claiming anything you cannot explain live.
Is the quantum predicate the same as the AC one?	No. The quantum predicate is the DC cascade, because only DC-type linear algebra could plausibly be computed coherently; AC is verification.	§3.2, E3	AC-only effects are outside the quantum oracle. Avoid: “The oracle is AC.”

25. Novelty and contribution

Class	Items
Standard quantum algorithms	Grover, BBHT, Dürr–Høyer, GAS, Dicke states (Bärtschi–Eidenbenz), QAOA with XY mixers, quantum counting
Standard power-flow modelling	DC/AC power flow, PTDF/LODF/Woodbury, quasi-steady-state cascade models
Standard contingency analysis	N-1/N-k screening, interdiction formulations
Standard optimization	MILP interdiction, simulated annealing, genetic algorithms, beam and hill-climbing
Novel combination	Cardinality-constrained amplitude amplification applied to a cascade predicate; DC search with AC verification in one reproducible pipeline
Engineering contribution	Oracle-agnostic pipeline; class-collapsed exact emulation at IEEE scale; claims lint and evidence ledger; interactive crossover explorer
Potential research contribution	(i) A component-level Toffoli model for a cascade oracle calibrated by a synthesized micro-oracle, including the injection-update term; (ii) a DC-versus-AC cascade ranking-agreement study with false-negative analysis under synthetic ratings; (iii) crossover analysis expressed through the structured-classical exponent α

No claim of being “first” is made. A literature search for prior quantum-search or Grover-based work on N-k contingency analysis must be completed before any such wording is used [PROPOSED].

Novelty statements

- **Abstract.** We formulate cardinality-constrained N-k cascading-outage search as amplitude amplification over Hamming-weight-k outage sets, build an open DC/AC quasi-steady-state cascade oracle, and give a component-level, micro-oracle-calibrated estimate of the oracle size, logical clock and classical-parallelism conditions under which a quadratic query reduction could become a wall-clock advantage. Against structured classical baselines on IEEE test systems we report [measured result].
- **Hackathon pitch.** We built a cascade simulator and searched it the way a quantum computer would — a real circuit on a toy grid, exact emulation on IEEE-118 — and then did the part most teams skip: we computed when it could actually win, and what has to be true.
- **Technical presentation.** Dicke-restricted Grover/Dürr–Høyer over an exactly-k outage space, verified at circuit level on toy instances and emulated exactly at IEEE scale; a synthesized micro-oracle calibrates a fault-tolerant resource model whose crossover is stress-tested against structured classical search.
- **Research paper.** We contribute (i) a reproducible DC/AC cascade-oracle benchmark for N-k search, (ii) an empirical DC–AC ranking-fidelity study under synthetic ratings, and (iii) a calibrated resource analysis showing the conditions, expressed through the classical scaling exponent α, under which quadratic speedups survive.

26. Roadmap

Phase	Dates	Deliverables	Done when
Round 1	through 7 Oct	IEEE-14 twin with DC cascade (AC if time allows); k = 2, 3 reference table; Tier B emulation and Dürr–Høyer; Tier A1 toy circuit on Aer; cost model and crossover figure; minimal cascade animation; 10-slide deck; 3-minute video with human voice-over; README. Anything unfinished is listed as Round 2 work, never shown as done.	Four submission items uploaded; Devfolio team exists
Round 2 sprint (a)	10–17 Oct	IEEE-30/118 DC tables; ratings calibration; E0–E3; baselines (E7)	E2, E3 tables filled from runs
Round 2 sprint (b)	18–25 Oct	E4–E6, E10–E12, E14, E15; A2 micro-oracle; noisy simulation; full web app	All experiments rerun from the Makefile
Freeze and rehearse	26–29 Oct	Final deck, video, README; claims lint passed; offline demo build; Q&A bank	Dry run under 7 minutes with no network dependency
Final / demo	30–31 Oct	In-person demo, pitch and jury Q&A	—
Future research	after the event	Real-grid data and ratings; dynamic validation; larger cases (e.g. Polish-2383) with measured structured-classical exponents; improved coherent oracles (incremental or approximate flow updates); synthesis of larger micro-oracles; literature verification of novelty	—

27. Conclusion

We do not claim that current quantum hardware can outperform classical contingency analysis on IEEE systems. Our own model indicates that, at IEEE scale, classical search wins, and the study is built so that this outcome is measured rather than assumed.

What Grid Doomsday Finder provides is a reproducible benchmark that (1) defines the exactly-k N-k cascading-outage search problem with explicit severity metrics and non-overclaiming terminology; (2) validates the power-flow and cascade machinery, and measures how far a DC screen can be trusted against AC verification; (3) implements and verifies the constrained quantum-search machinery — Dicke and ranked-index encodings, Grover, BBHT and Dürr–Høyer — at circuit level on toy instances and by exact emulation at IEEE scale; (4) synthesizes a toy-scale reversible cascade micro-oracle whose counted gate costs calibrate the resource model; (5) compares against strong, structure-exploiting classical methods under a fair protocol; and (6) quantitatively identifies — through oracle size, logical clock speed, classical parallelism and the classical scaling exponent — the conditions under which a future fault-tolerant quantum oracle could become competitive.

The scientific value lies in that map of conditions, and in a pipeline whose every claim carries an evidence label a reviewer can check.

Appendix A. Default parameters

Parameter	Default	Section
Rating headroom h	0.25 (calibrate within 0.15–0.40)	5.3
Stress factor λ	1.10 (calibrate within 1.0–1.3)	5.2
Floor s_{floor}	5% of largest base flow	5.3
Trip factor τ_{trip}	1.0	8.7
Max cascade rounds	30	8.7
Severity threshold τ	0.20 of base load (calibrate)	9
Cascade-driven threshold δ	0.10	9
Tie band	10^{-9}	8.4
AC tolerance	10^{-8} p.u.	7
Ladder step	5% (≤ 20 steps)	8.6
BBHT growth λ	6/5	12
Patience	3	12.2
GAS seeds	200	15.2
Fixed-point width b	16 (to be set by E12)	16
Cascade rounds in oracle R	4 (placeholder; E2)	16
k_{eff}	3k (placeholder; E2)	16
Logical Toffoli time t_T	10 µs (sweep 1–100 µs)	16
c_q	4 (placeholder; E6)	16
t_c	0.5 ms DC; 10 ms AC (placeholders; E1)	16
P	1 and 1000	16
QAOA layers p / shots	1–4 / 4096	14

Appendix B. Glossary

Term	Meaning
Bus / branch	Node / edge of the grid graph; branches are lines and transformers
N-k contingency	Simultaneous loss of k components
Cascade	Repeated cycle of island balance, power flow, overload and tripping until stable (here quasi-steady-state)
Island	Connected sub-network disconnected from the rest of the grid
PTDF / LODF	Power transfer / line outage distribution factors: linear flow sensitivities in DC
Severity S(x)	Fraction of base load unserved after the cascade settles
C(x)	$S(x) - S_0(x)$: cascade-induced loss
Oracle	Black-box predicate or function evaluated by the search (here, the cascade simulator)
Phase oracle	Gate flipping the sign of marked basis states
Diffuser	Reflection about the initial state: inversion about the mean amplitude
Dicke state	Equal superposition of all bitstrings with a given Hamming weight
BBHT	Boyer–Brassard–Høyer–Tapp: Grover search with unknown number of solutions
GAS-style threshold ascent	Repeated thresholded amplitude amplification raising the threshold on success (Dürr–Høyer; Gilliam et al.)
XY mixer	Weight-preserving QAOA mixer built from XX+YY interactions
Toffoli	Three-qubit controlled-controlled-NOT; the unit of fault-tolerant arithmetic cost
Crossover N*	Search-space size where quantum and classical runtimes are equal under the model
QROM	Quantum read-only memory: data lookup indexed by a quantum register

Appendix C. Derivations

C.1 Grover rotation

With $|a\rangle$, $|b\rangle$ the normalised marked and unmarked superpositions, $|s\rangle = \sin\theta|a\rangle + \cos\theta|b\rangle$ and $\sin^2\theta = M/N$. The oracle is the reflection $I - 2|a\rangle\langle a|$ and the diffuser is $2|s\rangle\langle s| - I$. The product of two reflections about axes at angle θ is a rotation by 2θ, so after r iterations the state is $\sin((2r+1)\theta)|a\rangle + \cos((2r+1)\theta)|b\rangle$ and $P = \sin^2((2r+1)\theta)$, maximal when $(2r+1)\theta \approx \pi/2$, i.e. $r \approx \pi/(4\theta) - ½ \approx (\pi/4)\sqrt{(N/M)}$ for $M \ll N$.

C.2 Closure of the Hamming-weight subspace

The oracle is diagonal in the computational basis and therefore maps each weight-k basis state to itself times ± 1. The reflection about $|D_k^L\rangle$ is $2|D\rangle\langle D| - I$, built as $U(2|1^k 0^{L-k}\rangle\langle\cdot| - I)U^\dagger$; since $|D\rangle \in H_k$, it maps H_k into itself. The algorithm therefore operates in an N-dimensional space in which $|D\rangle$ is the uniform vector, giving the amplitude update $\psi \mapsto 2\langle\psi\rangle - \psi$ used by the emulator.

C.3 DC compensation formula

With flat voltages and small angles, $P_{ij} = (\theta_i - \theta_j)/x_{ij}$, so bus balance gives $B'\theta = P$, $B' = A^T \operatorname{diag}(b) A$, singular until a slack is fixed. Removing branches K is equivalent to adding, at the terminals of each $m \in K$, a transfer z_m chosen so the flow on m becomes zero in the modified network. The flow change is $\Phi_{:,K}\, z$, and the condition $f'_K = f_K + \Phi_{K,K}\, z - z = 0$ gives $z = (I - \Phi_{K,K})^{-1} f_K$ (sign conventions as in §6.2), whence $f'_l = f_l + \Phi_{l,K}\, z$ for $l \notin K$. Test U4 confirms this against a direct re-solve.

C.4 Dürr–Høyer in brief

Choose a random threshold item; repeatedly run BBHT to find an item better than the current one; replace the threshold; stop after a budget proportional to $\sqrt{N}$. The expected number of improvements is logarithmic in N and the total expected number of oracle calls is $O(\sqrt{N})$, with the proven bound $22.5\sqrt{N} + 1.4 \lg^2 N$.

C.5 Class-collapsed Grover recursion

If amplitudes are α (marked) and β (unmarked) before an iteration, the oracle gives $(-\alpha, \beta)$; the mean is $\mu = (-M\alpha + (N-M)\beta)/N$; the diffuser gives $(2\mu + \alpha, 2\mu - \beta)$. This reproduces the full-vector update exactly, because the initial state and both operators are invariant under permutations within each class.

Appendix D. References

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